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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

There are more than 100 species of trees and shrubs around the world with the common name of 'ironwood'.

One type of ironwood tree, *Tabebuia avellanedae*, is native to South America.

Chemicals from these trees have antibacterial properties.

(a) (i) State the genus to which this plant belongs.

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(1)

.....

(ii) Which structure in the cells of this plant stores starch?

(1)

- ☒ **A** amyloplast
- ☒ **B** middle lamella
- ☒ **C** plasmodesmata
- ☒ **D** tonoplast

(b) Plague is a disease caused by *Yersinia pestis* bacteria.

Every year, thousands of people around the world are infected with these bacteria. These bacteria have a thick slime capsule and many pili.

(i) State the function of each of these structures.

(2)

Slime capsule .....

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Pili .....

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*(ii) The ironwood tree has wood containing chemical compounds called naphthoquinones. These compounds have antibacterial properties. Scientists are investigating whether these compounds could be used to develop a drug to treat people infected with *Y. pestis*. Describe the methods that would be used to trial a drug containing these compounds.

(6)

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(Total for question = 10 marks)

Q2.

Starch and cellulose are two biological molecules found in cells.

(a) Which type of biological molecule are starch and cellulose?

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(1)

- ☒ **A** phospholipid
- ☒ **B** polypeptide
- ☒ **C** polysaccharide
- ☒ **D** triglyceride

(b) Which structure stores starch in a plant cell?

(1)

- ☒ **A** amyloplast
- ☒ **B** mitochondrion
- ☒ **C** tonoplast
- ☒ **D** vacuole

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(c) Compare and contrast the structure of a cellulose molecule and a starch molecule.

(3)

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(Total for question = 5 marks)

Q3.

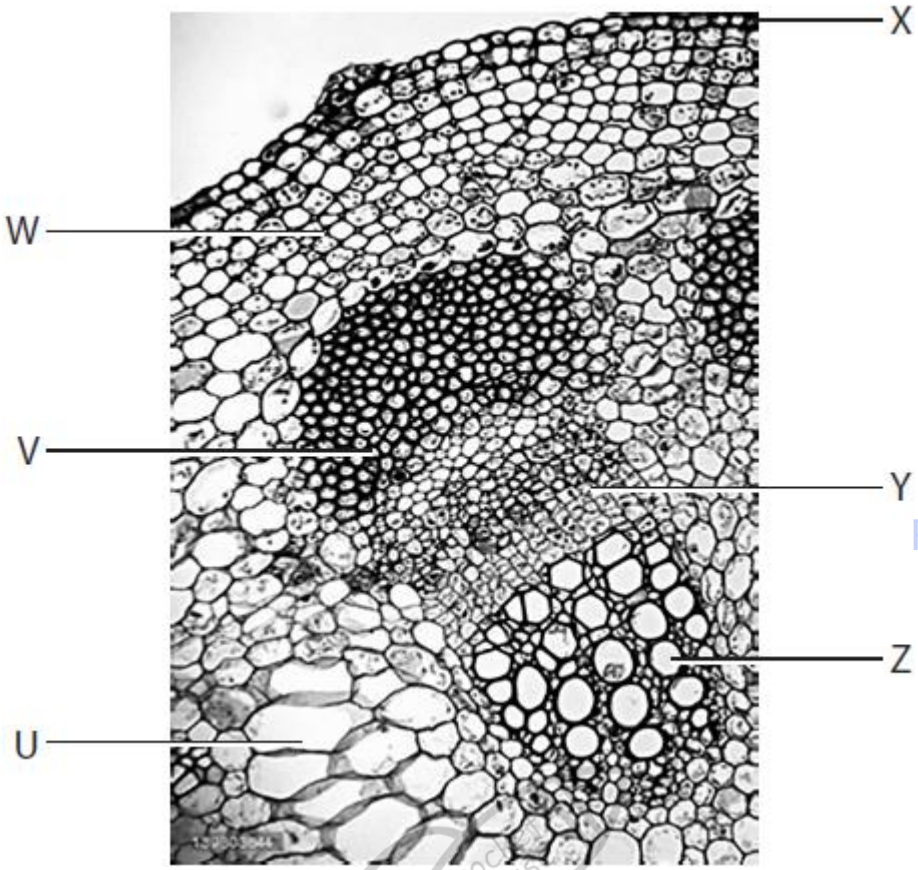
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Mineral ions are transported by xylem vessels.

The photograph shows a section through part of the stem of a plant, as seen using a light microscope.

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(a) (i) Which letter identifies a xylem vessel?

(1)

- ☐ **A** W
- ☐ **B** X
- ☐ **C** Y
- ☐ **D** Z

(ii) Which letter identifies a sclerenchyma fibre?

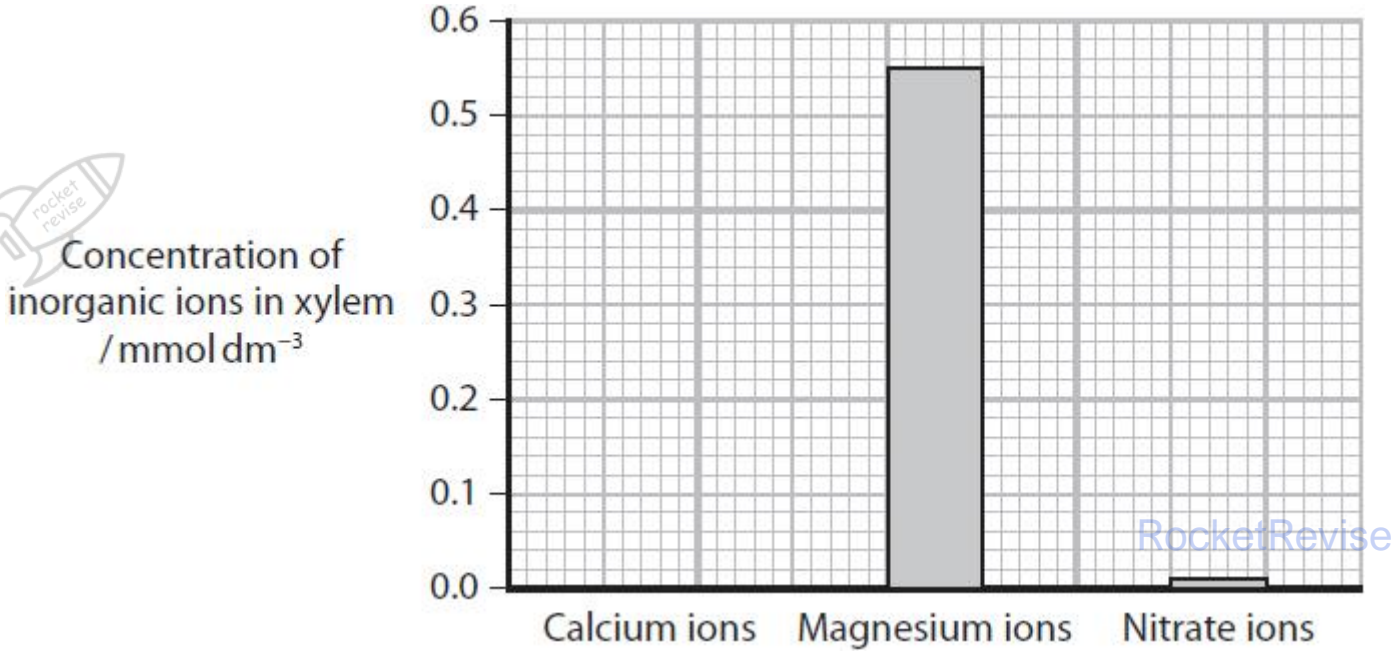
(1)

- ☐ **A** U
- ☐ **B** V
- ☐ **C** W
- ☐ **D** Z

(b) The graph shows the concentrations of three inorganic ions in the xylem of a plant.

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(i) The concentration of calcium ions is 0.45 mmol dm^{-3}.
Plot this concentration on the graph.

(1)

(ii) Explain the importance of each of these ions to the plant.

(3)

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(Total for question = 6 marks)

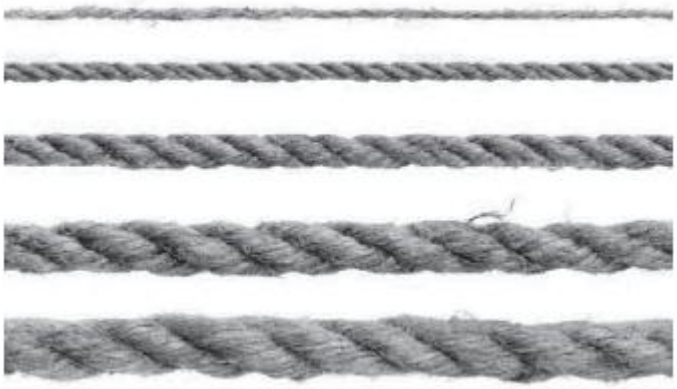
Q4.

Some plant fibres have a high tensile strength.

The photograph shows ropes made using fibres from the stems and leaves of hemp plants.

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Sclerenchyma fibres and xylem vessels are found in plant fibres.

(a) Both sclerenchyma fibres and xylem vessels contain lignin.

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(i) How many of the following statements are correct?

- Lignin adds strength to the cell membranes.
- Lignin gives waterproofing properties.
- Lignin can form spiral shapes.

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(ii) Give **one** similarity, other than the presence of lignin, and **one** difference in the structures of sclerenchyma fibres and xylem vessels.

(2)

Similarity

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Difference

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(b) Scientists have suggested that plant-based plastics should be used instead of oil-based plastics.

Hemp has been used to make some of these plant-based plastics.
Explain why hemp fibres are a sustainable resource.

(2)

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(c) Biofuels, produced from crop plants, are a sustainable resource.

Biofuels include bioethanol and biodiesel.
Bioethanol is produced from hemp and from sugar beet. Biodiesel is produced from rapeseed oil and from soybean oil.

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(i) The table shows information about the global production of bioethanol.

Year	Global production of bioethanol $\times 10^6$ dm^3
2000	17 000
2011	84 600

Determine what the global bioethanol production will be in 2020.
Assume the mean rate of increase stays constant.

(3)

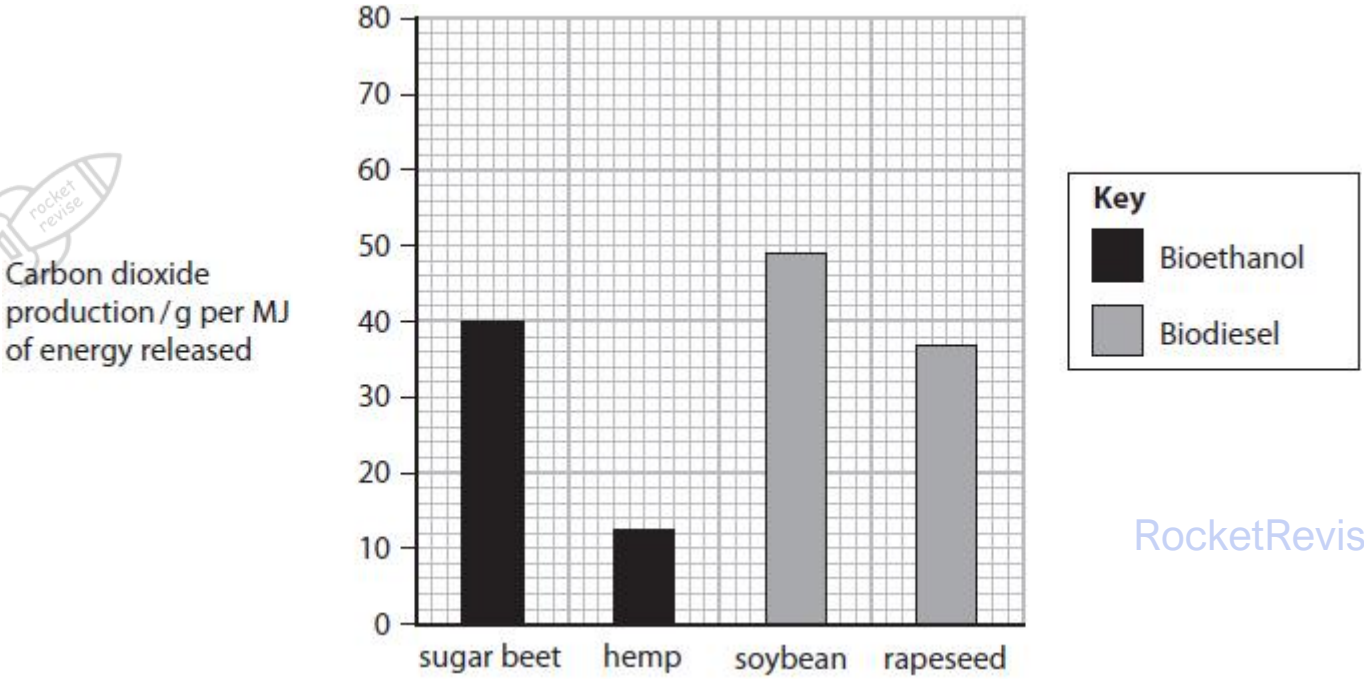
Answer .....

* (ii) The use of biofuels produces carbon dioxide. The graph shows the production of carbon dioxide using biofuels from four crop plants.

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The table gives information about the level of water, fertilisers and pesticides needed for crop plants to grow.

Crop plant	Level of resources used in production of biofuels		
	Water	Fertilisers	Pesticides
Hemp	Medium to low	Low	Low
Rapeseed	High	Medium	Medium
Soybean	High	Low to medium	Medium
Sugar beet	Medium to low	High	Medium

Explain the advantages and disadvantages of using these crop plants as a sustainable resource to produce biofuels.
Use the information in the question to support your answer.

(6)

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(Total for question = 14 marks)

Q5.

The photograph shows a Venus fly trap (*Dionaea muscipula*), a plant native to North and South Carolina in the United States.



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This plant is found in nitrogen and phosphorus poor environments, such as a bog. The plant catches and digests insects and spiders and absorbs compounds containing nitrogen.

(a) Explain why the Venus fly trap needs compounds containing nitrogen.

(2)

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(b) The Venus fly trap secretes extracellular enzymes that digest the insects and spiders it catches.

Describe the role of the Golgi apparatus in forming these extracellular enzymes.

(2)

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(c) Extracts from Venus fly trap plants are used in some alternative medicines as a treatment for HIV and some cancers.

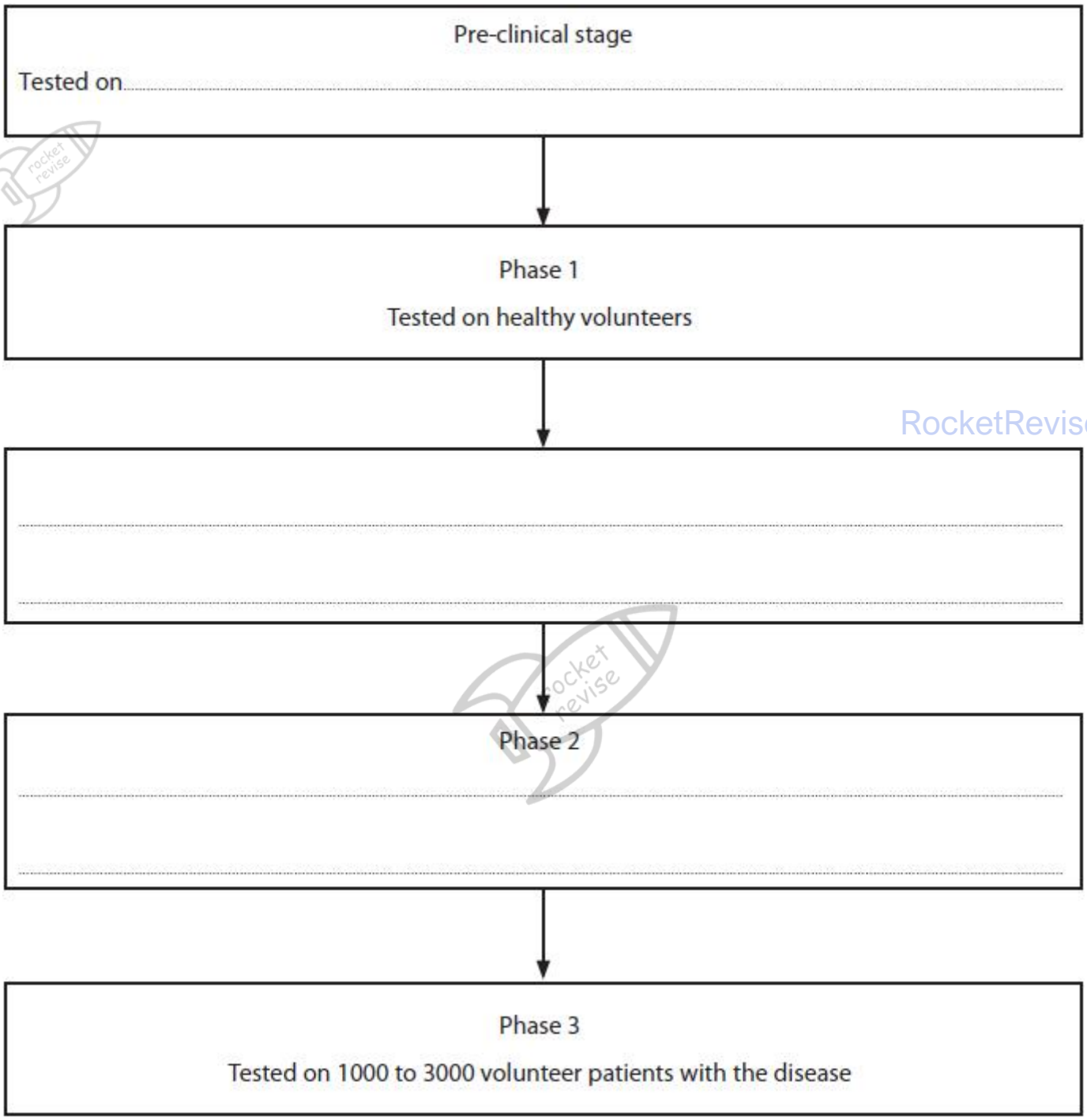
Complete the diagram to show how an extract could be tested as part of the initial stages of a drug trial.

(3)

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(Total for question = 7 marks)

Q6.

All organisms contain one or more cells.

(a) (i) In which of the following is a cell membrane present?

(1)

- ☒ **A** animal cells only
- ☒ **B** animal and plant cells only

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- ☒ **C** plant and prokaryotic cells only
- ☒ **D** animal, plant and prokaryotic cells

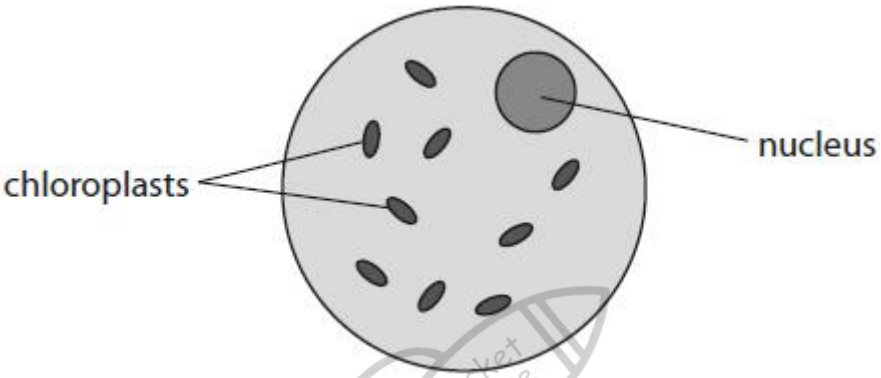
(ii) In which of the following is a cell wall present?

(1)

- ☒ **A** plant cells only
- ☒ **B** animal and plant cells only
- ☒ **C** plant and prokaryotic cells only
- ☒ **D** animal, plant and prokaryotic cells

(b) Sailor's eyeball (*Valonia ventricosa*) is a single-celled, spherical organism.

The diagram shows one of these cells.



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(i) The diameter of this cell is 25 μm.
Calculate the magnification of the diagram.

(2)

Answer .....

(ii) This cell contains chloroplasts.
State the function of these chloroplasts.

(1)

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(iii) This cell is not a prokaryotic cell as it contains chloroplasts.
Give **one** other reason why this organism is not a prokaryotic cell.

(1)

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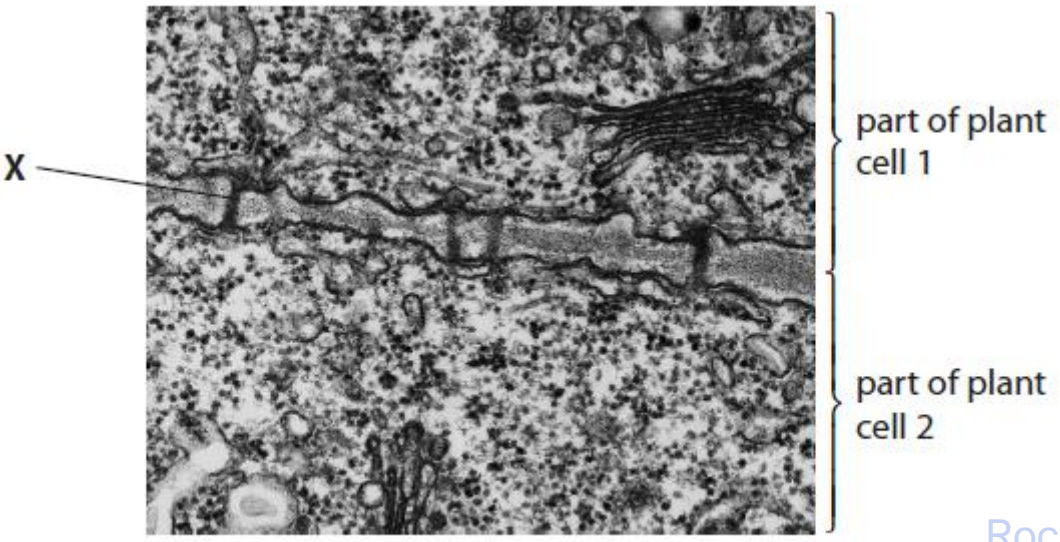
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(c) The photograph shows part of two adjoining plant cells, as seen using an electron microscope.

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(Source: © biophoto associates/Getty Images)

(i) Name the part labelled **X**.

(1)

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(ii) Explain the function of the part labelled **X**.

(2)

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(Total for question = 9 marks)

Q7.

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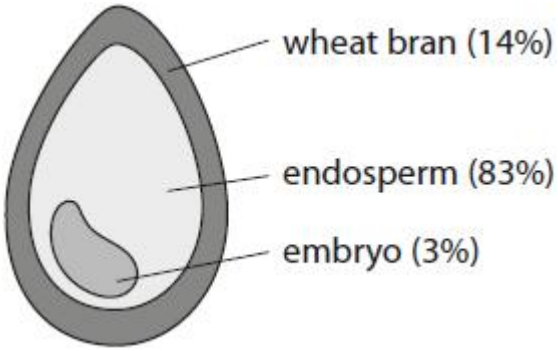
Plant-based products can replace plastics produced from oil.

Wheat bran comes from the outer layers of grains of wheat.

(a) The diagram shows the structure and composition of a grain of wheat.

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Wheat bran contains 43% fibre.
A grain of wheat has a mass of 48 mg.
Calculate the mass of fibre in the wheat bran of this grain of wheat.
Give your answer to **two** significant figures.

(3)

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Answer ..... mg

(b) Disposable plates and cutlery can be made from either plant-based products or from oil-based plastics.

The photograph shows cutlery and a plate made from wheat bran.



(Source: © Konektus Photo/Shutterstock)

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Explain the advantages of using cutlery and plates made from wheat bran instead of oil-based plastics.

(3)

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(c) The endosperm of a grain of wheat develops from the endosperm nucleus, formed during fertilisation.

Explain the role of the pollen tube and nuclei in the formation of the endosperm nucleus.

(3)

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(Total for question = 9 marks)

Q8.

The photograph shows kenaf plants growing on a farm in Nigeria.



(Source: © Design Pics Inc/Alamy Stock Photo)

Kenaf is grown mainly in parts of Africa and southern Asia.

Fibres from the stems of these plants are used to make paper.

Fibres are obtained from the xylem and the phloem.

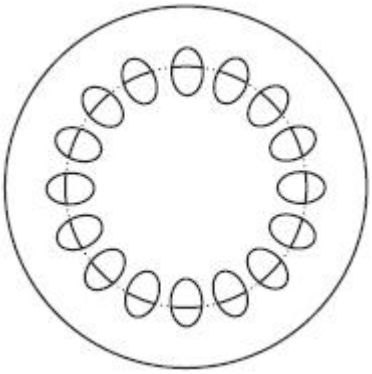
(a) (i) The diagram shows a transverse section through the stem of a plant.

Label the diagram to show the position of the xylem and the phloem.

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(1)



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(ii) Below are some statements about xylem and phloem.

- both tissues have a transport function
- xylem vessels contain sieve plates
- phloem transports polysaccharides
- xylem vessels can transport substances in both directions

How many of these statements are correct?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(b) When a plant is not given sufficient water for a period of time, it develops water stress.
In an investigation, three rows of kenaf plants were grown in separate containers in a glasshouse.
The temperature, light intensity, soil quality and soil quantity were controlled.
The diagram shows how each row was treated.

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Row 1 → Plants given 100 cm^3 of water every day

Row 2 → Plants given 100 cm^3 of water every day for 1 month and then 10 cm^3 of water every day

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Row 3 → Plants given 100 cm^3 of water every day until flowering (after 2 months) and then 10 cm^3 of water every day

The table shows the means and standard deviations of the measurements of the fibres extracted from the plants at the end of this investigation.

Row	Mean length / mm	Mean diameter / µm	Mean cell wall thickness / µm
1	0.91 ± 0.01	26.95 ± 0.45	3.44 ± 0.16
2	0.64 ± 0.06	21.62 ± 0.71	2.85 ± 0.08
3	0.81 ± 0.03	26.48 ± 0.77	3.23 ± 0.09

(i) Calculate the percentage decrease in the mean diameter of the fibres from plants in row 2 compared with those from plants in row 1.

(3)

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Answer ..... %

(ii) Comment on the results of this investigation.
Use the information in the table to support your answer.

(4)

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(c) The walls of xylem vessels contain cellulose.

Which of the following shows the structure of part of a cellulose molecule?

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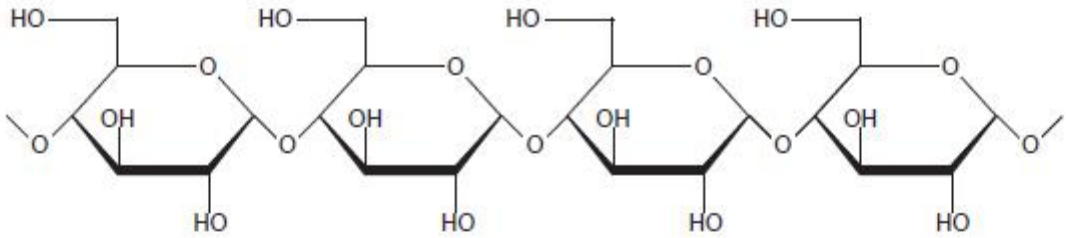
(1)

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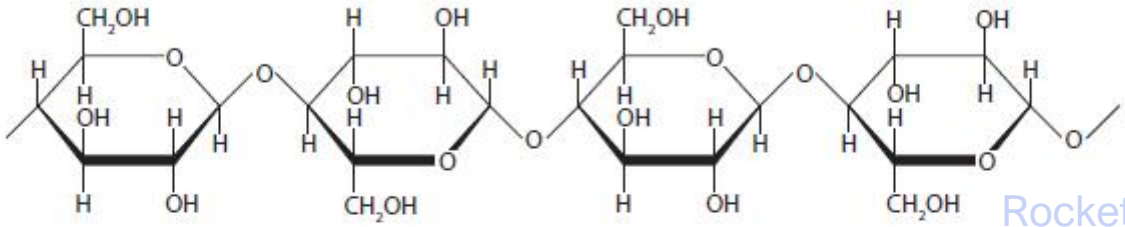
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☒ A

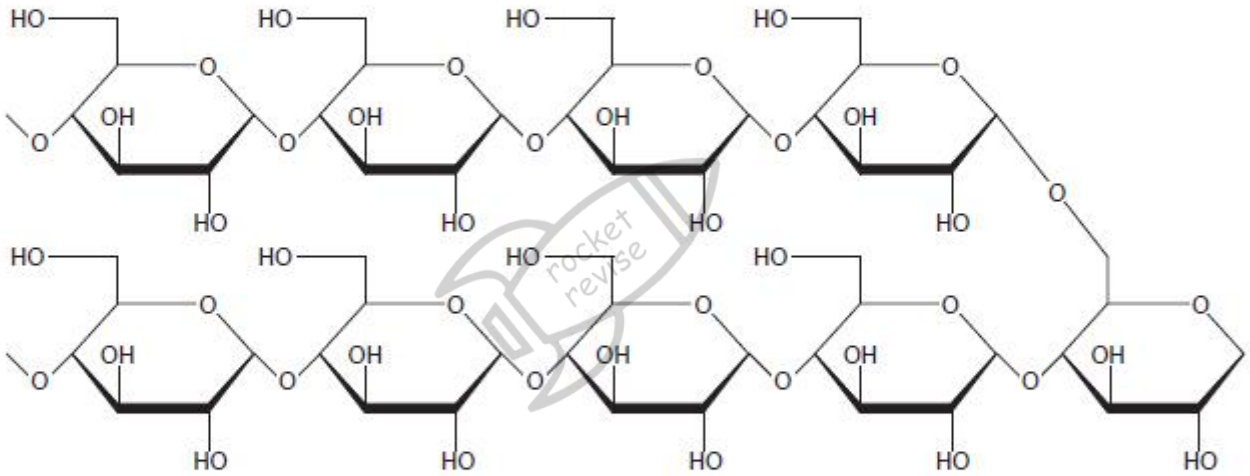


☒ B

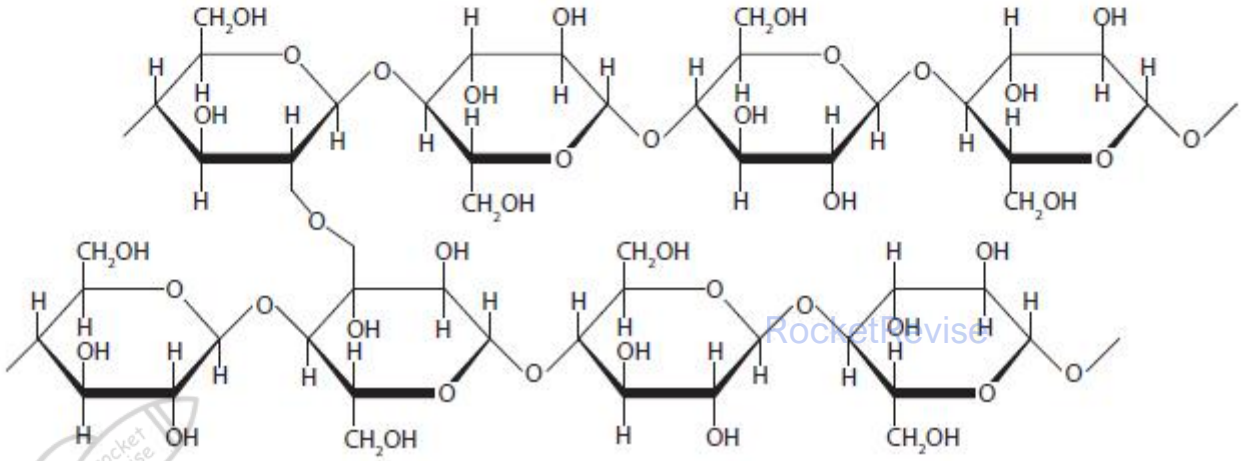


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☒ C



☒ D



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(d) Explain how xylem vessels are adapted for support and transport.

(4)

Support

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Transport

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(Total for question = 14 marks)

Q9.

Plants contain starch and cellulose.

(a) (i) Which part of a plant cell stores starch?

(1)

- ☒ **A** amyloplast
- ☒ **B** middle lamella
- ☒ **C** plasmodesmata
- ☒ **D** tonoplast

(ii) How many of the following statements about starch are correct?

1. it has a compact shape
2. it contains 1,6 glycosidic bonds only
3. it is a polymer of β-glucose
4. it is a polypeptide

(1)

- ☒ **A** one
- ☒ **B** two
- ☒ **C** three
- ☒ **D** four

(iii) How many of the following contain cellulose?

1. phloem
2. sclerenchyma
3. vacuole
4. xylem

(1)

- ☒ **A** one
- ☒ **B** two
- ☒ **C** three

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☒ **D** four

(b) Microfibrils are composed of cellulose.

The photograph shows the arrangement of microfibrils in a plant cell wall, as seen using an electron microscope.



(Source: © Biophoto associates/Science photo library)

Explain how the structures of cellulose and microfibrils increase the strength of a plant cell wall. Use the information in the photograph to support your answer.

(3)

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(Total for question = 6 marks)

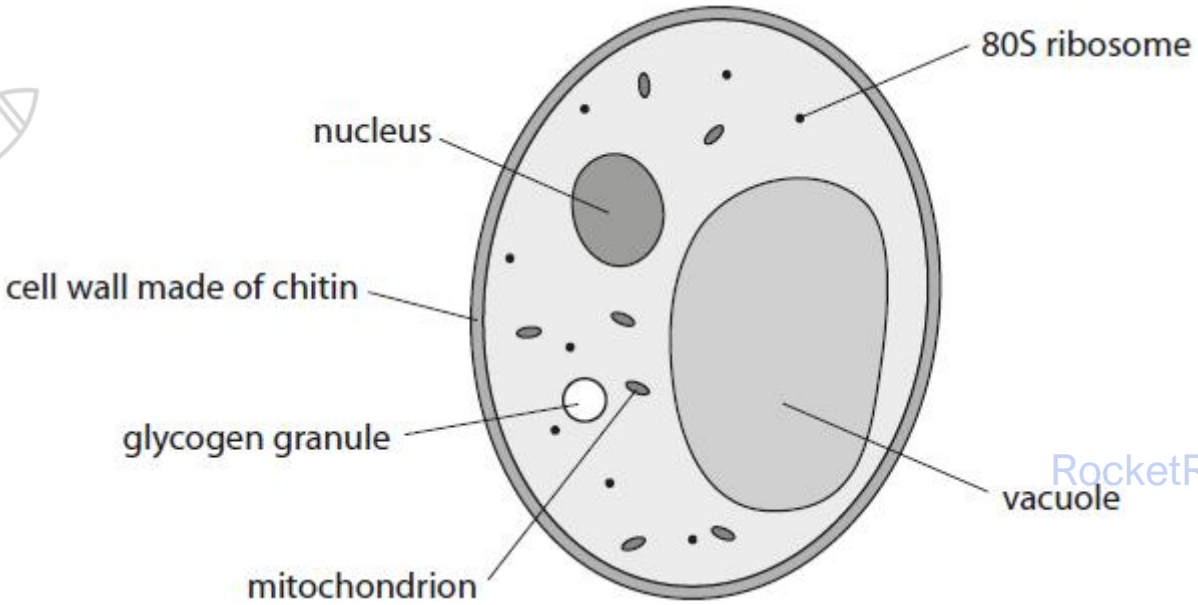
Q10.

Some fungi infect plants and affect the expression of the plant genes.

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(a) A student drew and labelled a fungal cell.



Compare and contrast the structure of this fungal cell and a plant cell.

(3)

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(b) The photograph shows darnel, a species of grass.

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© blickwinkel / Alamy Stock Photo

The fungus *Epichloë festucae* lives within darnel for part of its life cycle.
The fungus influenced the expression of certain genes in the plant cells.

- There was reduced expression of some genes involved in DNA synthesis.
- There was reduced expression of some genes involved with the synthesis of phospholipids, starch and sucrose.

(i) Explain the effect of this fungal infection on the growth of the plant.

(4)

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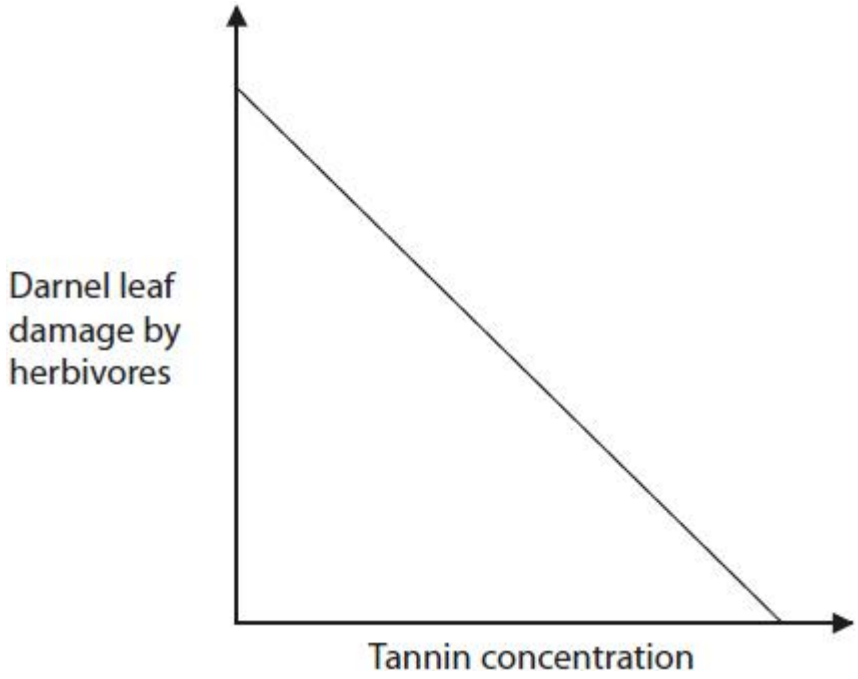
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*(ii) The fungus and the plant both benefit from this relationship.
The fungus absorbs nutrients from the plant.
The expression of certain genes in the plant cells is increased when infected with the fungus.
These genes are involved in the synthesis of tannin and flavonoids.
The graph shows the relationship between tannin concentration and the degree of darnel leaf

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damage by herbivores (grazing animals).



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The table shows the antimicrobial effect of different concentrations of flavonoids on cultures of the bacterium *Pseudomonas aeruginosa*. This bacterium causes disease in plants.

Flavonoid concentration / $\mu g\ cm^{-3}$	Diameter of inhibition zone / mm
500	7.0
666	8.0
1000	9.0

The fungus increases the expression of the genes involved in the synthesis of tannin and flavonoids. Discuss the advantages and disadvantages of this for the fungus and for the infected plants.

(6)

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(iii) The bacterium *Pseudomonas aeruginosa* causes lung infections in humans. Describe how a drug containing flavonoids could be tested in a stage II drug trial.

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(Total for question = 15 marks)

Q11.

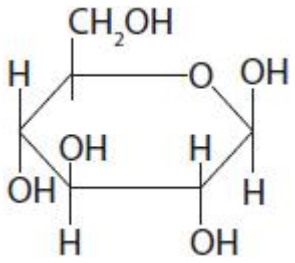
Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Plants contain many types of molecule.

(a) (i) Which molecule is shown in the diagram?

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(1)

- ☒ **A** *α*-glucose
- ☒ **B** *β*-glucose
- ☒ **C** starch
- ☒ **D** sucrose

(ii) Which is a feature of cellulose molecules?

(1)

- ☒ **A** contain peptide bonds
- ☒ **B** contain *α*-glucose molecules
- ☒ **C** form microfibrils
- ☒ **D** are soluble

(iii) Which molecule contains magnesium ions?

(1)

- ☒ **A** calcium pectate
- ☒ **B** chlorophyll
- ☒ **C** DNA
- ☒ **D** starch

(b) Plants contain tissues that have different functions.

(i) Name **one** tissue that gives the plant support and is also involved in the transport of water through the plant.

(1)

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(ii) Name **one** fibre that gives the plant support but is **not** involved in the transport of water through the plant.

(1)

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(Total for question = 5 marks)

Q12.

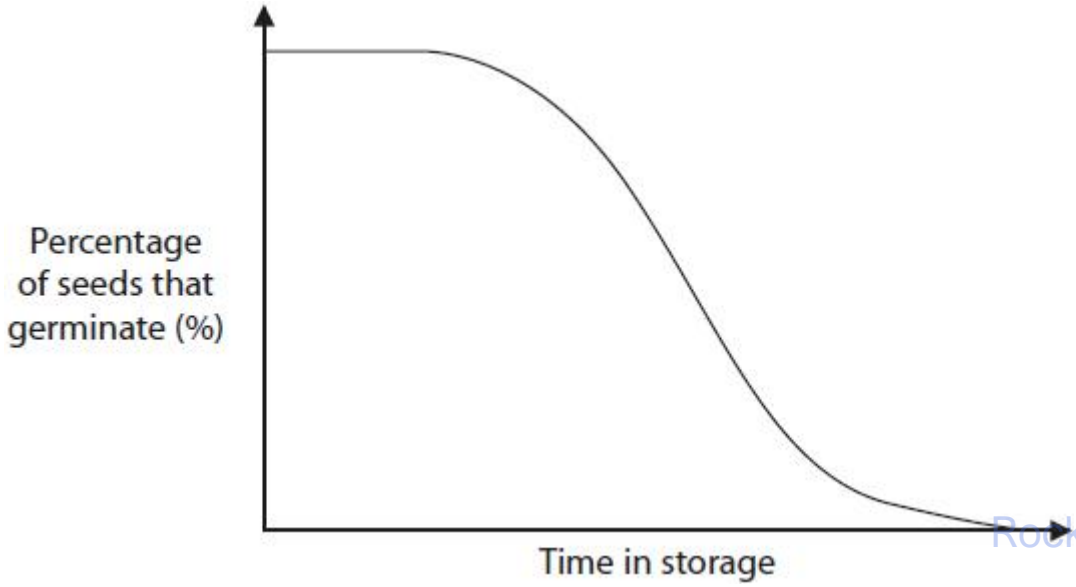
Seed banks are used to store seeds from a wide variety of plants. Some plants are a sustainable source of bioplastics.

(a) The Germplasm Bank of Wild Species is the largest seed bank for wild species of plants in Asia.

Seeds from 8 500 species have been collected and stored.
Seeds were dried before being stored at temperatures below 0 °C.
The effect of length of time in storage on seed germination is shown in the graph.

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(i) Describe the effect of length of time in storage on the germination of these seeds.

(2)

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(ii) Two of the methods used in seed banks are:

- store seeds from many different plants of the same species, instead of many seeds from just one plant
- regularly germinate samples of the stored seeds, allowing them to grow into adult plants.

Explain the advantages of these methods.

(4)

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(b) The photograph shows bioplastic straws and cutlery, produced from the seeds of avocados.



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© Facinadora / Alamy Stock Photo

(i) In 2018, a company in Mexico produced 130 000 kg of bioplastic cutlery and straws per month. 40% of the products were straws.
Calculate the mass of cutlery produced per year by this company.
Give your answer to **two** significant figures.

(2)

Answer .....

(ii) The use of these plant-based products is more sustainable than the use of cutlery and straws made from oil-based plastic.
Explain what is meant by the term sustainable, with reference to the cutlery produced from the seeds of avocados.

(2)

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(Total for question = 10 marks)

Q13.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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A coral reef is an underwater ecosystem.

The photograph shows a coral reef in the Caribbean.



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(Source: © Belize / Alamy Stock Photo)

A photosynthetic bacterium found on these coral reefs produces a chemical called curacin A.

This chemical has been found to be effective against some colon and kidney cancers.

(a) Large quantities of this organism need to be grown in a culture medium in order to extract enough curacin A.

The culture is kept at the optimum temperature for the growth of this bacterium.
Suggest **two** other conditions that would be needed for maximum growth of this bacterium.

(2)

1 .....

2 .....

(b) Scientists conduct experiments to determine the effectiveness and safety of new drugs.

(i) Which of the following scientists tested the effectiveness of digitalis soup?

(1)

- ☒ **A** Adolf Fick
- ☒ **B** Matthew Meselson and Franklin Stahl
- ☒ **C** Godfrey Hardy and Wilhelm Weinberg
- ☒ **D** William Withering

(ii) Curacin A is being developed as a drug to treat some colon and kidney cancers in humans.
Suggest how a suitable dose for cancer treatment would be determined in human trials.

(2)

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(iii) Describe how a double-blind clinical trial would be performed with this cancer drug.

(4)

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(Total for question = 9 marks)

Q14.

Stomach ulcers are sores that develop in the lining of the stomach.

The mucus produced by the stomach prevents bacteria from reaching the lining of the stomach.

Stomach ulcers can be caused by infection with the bacterium *Helicobacter pylori*.

These bacteria reproduce in the stomach as it provides all the conditions required for the growth of bacteria.

(a) Explain the conditions needed for the growth of bacteria.

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(b) The plant *Cibotium barometz* is native to several countries in East Asia.

The photograph shows hairs on the surface of this plant.



(Source: © Mauro Rodrigues/123RF)

Extracts from the hairs on this plant have antimicrobial properties.
An investigation compared the use of this extract and the drug omeprazole to treat stomach ulcers in rats.
The mass of mucus and the area of ulcer were measured in five groups of rats.
The table shows the treatments and the results of this investigation.

Group	Treatment / mg kg^{-1}	Mean mass of mucus / g	Mean area of ulcer / mm^2
Control rats with no ulcer	0.00	2.28	0.00
Control rats with ulcer	0.00	0.75	802.71
Rats with ulcer treated with omeprazole	20.00	1.92	95.71
Rats with ulcer treated with extract	125.00	1.87	202.80
Rats with ulcer treated with extract	500.00	2.17	172.80

(Source: adapted from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5244617/)

(i) Calculate the percentage decrease in the mean areas of ulcer of the rats treated with omeprazole compared with
the control rats with ulcer.

(2)

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Answer ..... %

*(ii) Explain the results of this investigation.
Use all the information in Question 6 to support your answer.

(6)

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(Total for question = 12 marks)

Q15.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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Brazil is the largest producer of sugarcane.

The photograph shows sugarcane plants growing in Brazil.



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(Source: © jeep2499/Shutterstock)

(a) Products from sugarcane, such as starch and cellulose, are used to make bioplastic.

(i) How many of the following statements about **starch** are correct?

- contains α-glucose and β-glucose molecules
- contains amylose and amylopectin
- stored in the tonoplast

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(ii) How many of the following statements about a **cellulose** molecule are correct?

- contains α-glucose and β-glucose molecules
- contains 1,4 glycosidic bonds
- forms hydrogen bonds with other cellulose molecules

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

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(b) Bioplastic bags made from sugarcane take up to six months to decompose.

Bacteria secrete enzymes onto the bags during the process of decomposition.

(i) Name **two** structures found in bacterial cells that are involved in the synthesis of these enzymes.

(1)

1 .....

2 .....

(ii) Decomposition of bioplastic bags occurs faster if there is increased bacterial growth.
Explain the conditions needed for increased bacterial growth.

(3)

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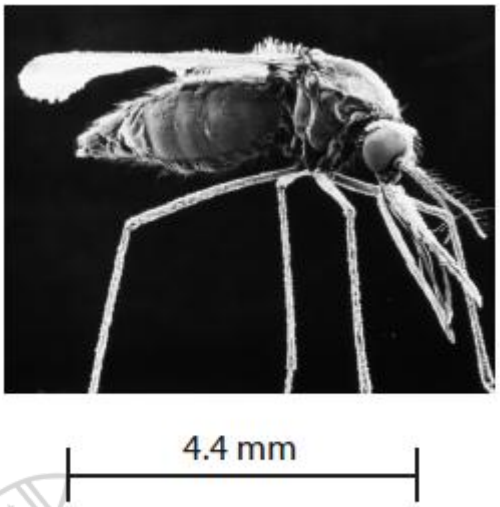
(Total for question = 6 marks)

Q16.

Female mosquitoes feed on the blood of other animals to promote the development of their eggs. During this process, infected mosquitoes can transmit the *Plasmodium* parasite.

The *Plasmodium* parasite causes the disease malaria.

The photograph shows a female *Anopheles* mosquito.



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(Source: © Dr Tony Brain / Science Photo Library)

(a) Calculate the magnification of the *Anopheles* mosquito in this photograph.

Give your answer to one decimal place.

(1)

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Answer .....

(b) In 2017, an estimated 219 million people in 87 countries had malaria, and 435 000 of these people died.

Which of the following shows the percentage of infected people who died?

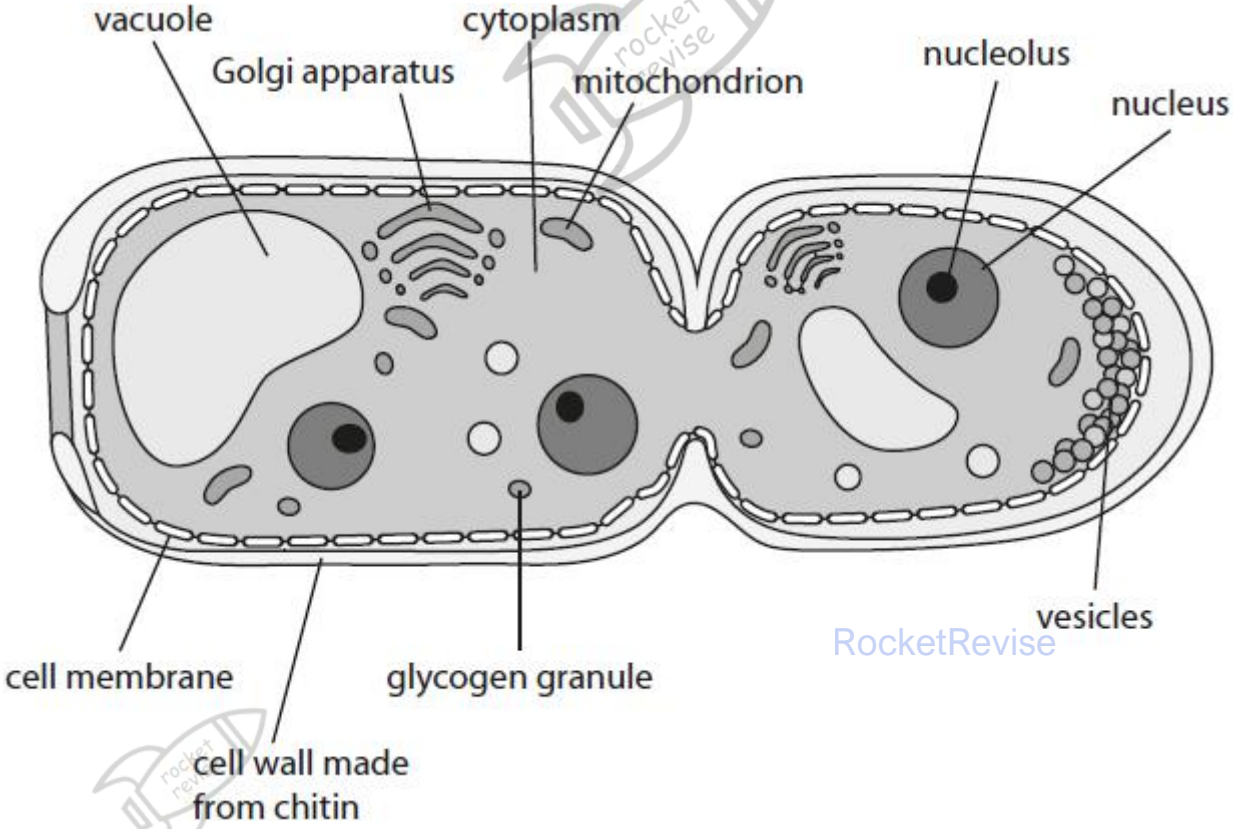
(1)

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- ☐ **A** 0.02%
- ☐ **B** 0.20%
- ☐ **C** 99.80%
- ☐ **D** 99.98%

(c) The fungus *Metarhizium* pingshaense infects *Anopheles* mosquitoes.

A student drew a labelled diagram of a fungal cell that was dividing.



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(i) State the function of a nucleolus.

(1)

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(ii) Give **three** differences in ultrastructure between this cell and a cell taken from the root of a plant.

(3)

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*(d) Insecticides are chemicals used to kill insects.

Anopheles mosquitoes have developed resistance to the insecticides currently used.
Metarhizium pingshaense is a fungus that infects *Anopheles* mosquitoes.
Scientists in West Africa have genetically engineered (GE) this fungus using a gene from a spider.
When the GE fungus infects an *Anopheles* mosquito this gene becomes activated. A toxin is released, which kills the mosquito.
In an investigation, four huts each containing a group of 500 breeding pairs of mosquitoes were used.
Three different treatments used to kill mosquitoes were compared with a control group.
The walls and ceiling of the huts were sprayed with the treatment.



(Source: © Ivan Vdovin / Alamy Stock Photo)

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The table shows the number of adult mosquitoes in the first two generations of offspring from each hut after spraying.

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Treatment	Number of adult mosquitoes in each generation	
	First	Second
Control (no treatment)	921	1396
Insecticide	919	1353
Normal fungus	436	455
GE fungus	399	13

Evaluate these three treatments.
Use the information in the question, and your own knowledge, to support your answer.

(6)

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(Total for question = 12 marks)

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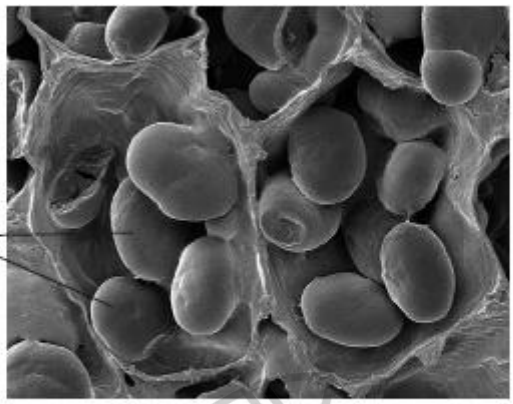
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Q17.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The cassava root is an important food crop grown in the tropics.

The photograph shows organelles containing starch granules inside cassava root cells, as seen using an electron microscope.



(Source: © Science Photo Library / Alamy Stock Photo)

(a) (i) Name these organelles found in cassava root cells.

(1)

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(ii) Explain how the structure of starch relates to its function.

(3)

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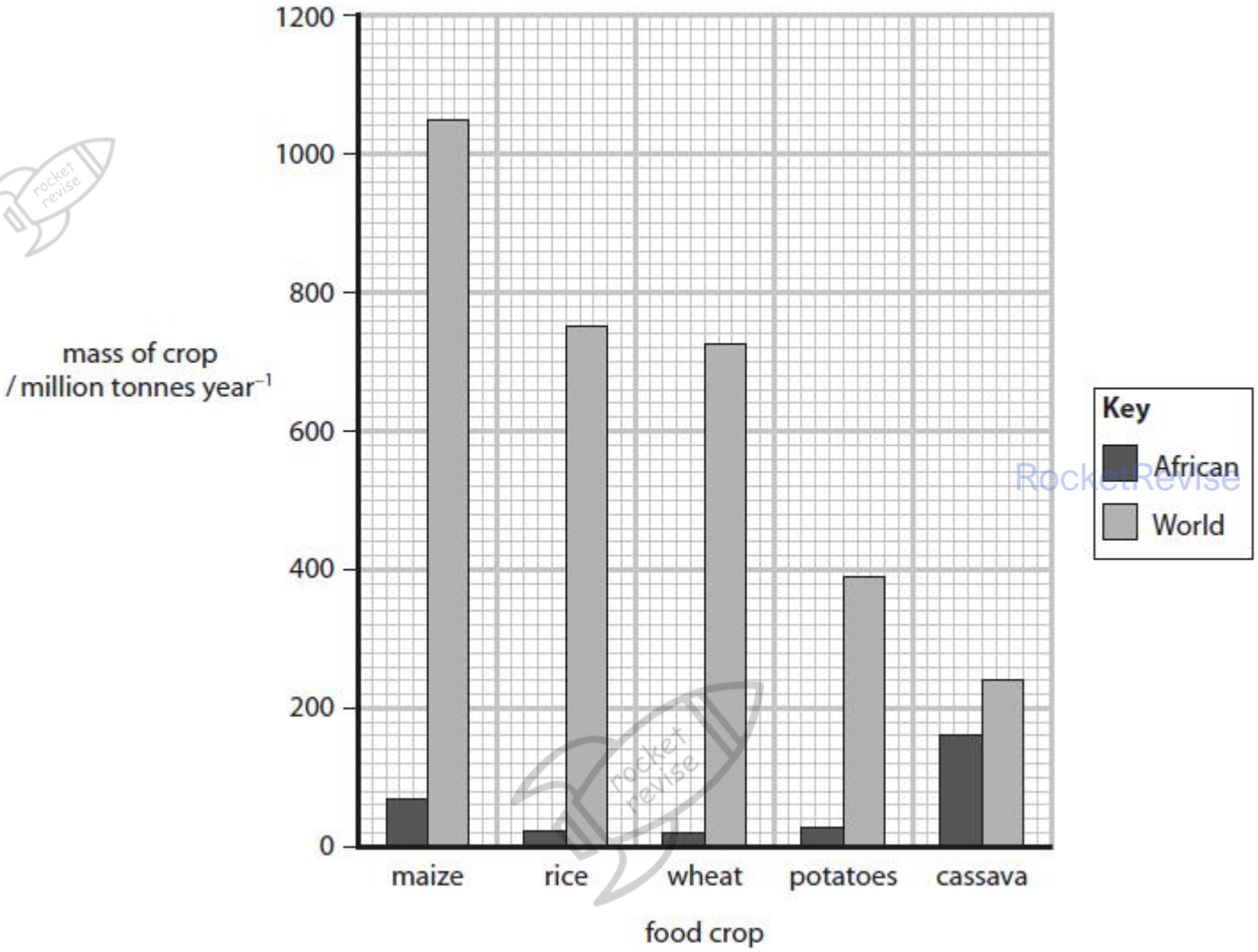
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(b) The graph shows the mass of five food crops grown in Africa. It also shows the world production of these crops.

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(i) Which is the percentage of the world cassava crop that is grown in Africa?

(1)

- ☒ **A** 33 %
- ☒ **B** 34 %
- ☒ **C** 66 %
- ☒ **D** 67 %

(ii) It is predicted that the world maize production will increase by 2.3 % in the following year.
Calculate the world production of maize in the following year.
Give your answer to the nearest whole number.

(1)

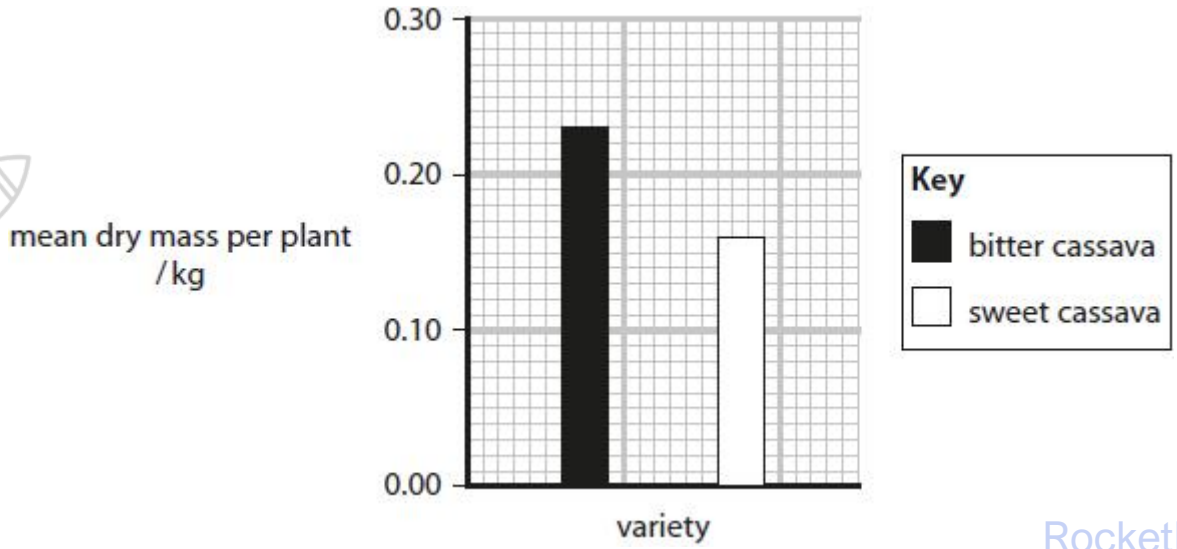
Answer ..... million tonnes

* (c) Some plants contain chemicals that protect them from being eaten by animals.

These chemicals taste bitter and can be converted by enzymes into hydrogen cyanide.
Hydrogen cyanide inhibits respiration and can cause death.
Cassava can be grouped into two main varieties: bitter cassava and sweet cassava. These varieties differ in the concentration of the bitter-tasting chemicals.
Processing of the cassava removes most of these chemicals.
The graph shows the mean dry mass of these cassava varieties when grown in the same conditions.

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The table gives information about three crop plants.

Crop plant	Water requirements	Minimum level of nutrients required in soil	Typical mass of bitter tasting chemicals / mg kg^{-1}	Typical mass of carbohydrate / 100 g of dry plant matter	Typical mass of protein / 100 g of dry plant matter
Bitter cassava	low to medium	low	100 to 500	94.70	1.80
Sweet cassava	low to medium	low	10 to 50	94.62	1.84
Maize	medium to high	medium to high	1 to 2	77.46	8.75

Discuss the advantages and disadvantages of growing these three crop plants for human food.
Use the information in the question to support your answer.

(6)

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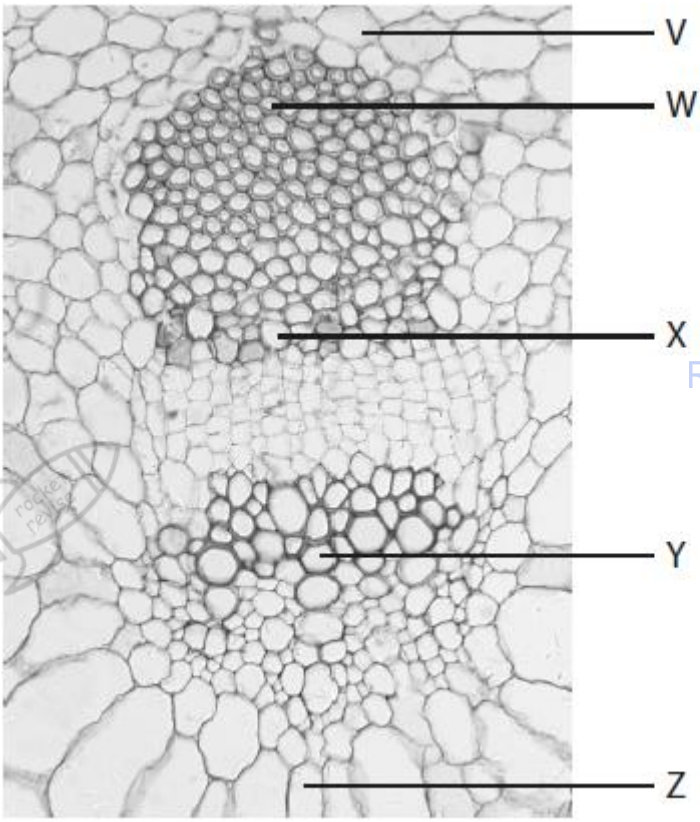
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(Total for question = 12 marks)

Q18.

The stem of a plant contains structures involved in transporting substances and supporting the plant.

The photograph shows a section through part of the stem of a plant, as seen using a light microscope.



(Source: © Choksawatdikorn/Shutterstock)

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(a) (i) Which of the labelled structures is sclerenchyma?

(1)

- ☒ **A** W
- ☒ **B** X
- ☒ **C** Y
- ☒ **D** Z

(ii) Which of the labelled structures is involved in transporting substances both up and down the plant stem?

(1)

- ☒ **A** W
- ☒ **B** X
- ☒ **C** Y
- ☒ **D** Z

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(iii) Which of the labelled structures is involved in transporting calcium ions from the stem to the leaves?

(1)

- ☒ **A** V
- ☒ **B** W
- ☒ **C** X
- ☒ **D** Y

(iv) Which of the following may have secondary thickening in the cell walls?

(1)

- ☒ **A** phloem sieve tubes only
- ☒ **B** phloem sieve tubes and sclerenchyma fibres
- ☒ **C** xylem vessels only
- ☒ **D** xylem vessels and sclerenchyma fibres

(b) Plants require inorganic ions for growth.

The photograph shows a leaf with magnesium ion deficiency.

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(Source: © Nigel Cattlin / Science Source / Science Photo Library)

In an investigation, groups of plants were watered with the same volume of a nutrient solution.
The solution contained all the required inorganic ions, but with different concentrations of magnesium ions.
The table shows the results of this investigation.

Concentration of magnesium ions in nutrient solution / mmol dm^{-3}	Mean dry mass of root / g	Mean dry mass of shoot / g	Mean total dry mass / g
0.01	2.38 ± 0.15	15.56 ± 1.22	17.94 ± 1.38
0.10	5.42 ± 0.43	33.36 ± 1.21	38.78 ± 1.63
0.40	7.08 ± 0.50	46.06 ± 0.59	53.14 ± 0.50

(i) Explain the importance of magnesium ions to plants.

(2)

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(ii) Comment on the results of this investigation.

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Use the information in the table to support your answer.

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(Total for question = 10 marks)

Q19.

Sweet pea plants were originally found in Sicily and southern Italy.

They are used to investigate genetic inheritance and linkage.

The photograph shows a flowering sweet pea plant.

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(Source: © Tim Gainey/Alamy Stock Photo)

Sweet pea plant flowers can be different colours and have a variety of pollen grain shapes.

(a) (i) How many of the following statements about a pollen grain are correct?

- contains a tube nucleus and a generative nucleus
- the generative nucleus divides by mitosis to form two female gametes
- can produce diploid gametes

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(ii) The table gives information about the surface area and volume of round and long pollen grains.

Pollen grain shape	Surface area	Volume / μm^3	Surface area to volume ratio
round	5.03	33.5	0.150
long		25.8	0.160

Calculate the difference in surface area between round and long pollen grains.
Give your answer in standard form with appropriate units.

(2)

Answer .....

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(b) Plants that grow from a sweet pea seed contain stem cells.

Describe how a stem cell in a plant can become a sclerenchyma cell.

(4)

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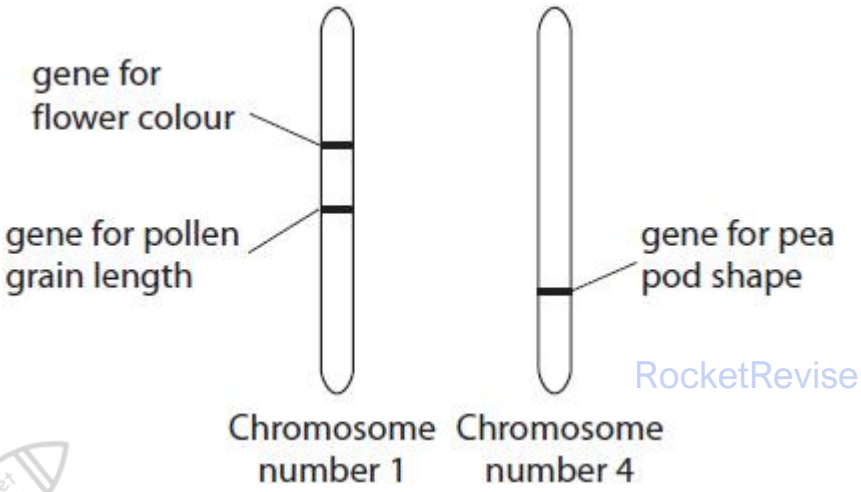
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(c) Sweet pea plants were used in an investigation into the inheritance of flower colour, pollen grain length and pea pod shape.

Each of these traits is controlled by a single gene. Sweet peas have seven pairs of chromosomes. The arrangement of these three genes on sweet pea chromosomes is shown in the diagram.



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Comment on the role of meiosis in the inheritance of these traits.

(4)

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(Total for question = 11 marks)

Q20.

Answer the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows water lilies in a river in Thailand.



(Source: © Igor Prahin/Alamy Stock Photo)

(a) One water lily leaf has a diameter of 305 cm.

Which of the following shows the upper surface area of this leaf?
Assume the leaf is circular.
Use the formula πr^2.

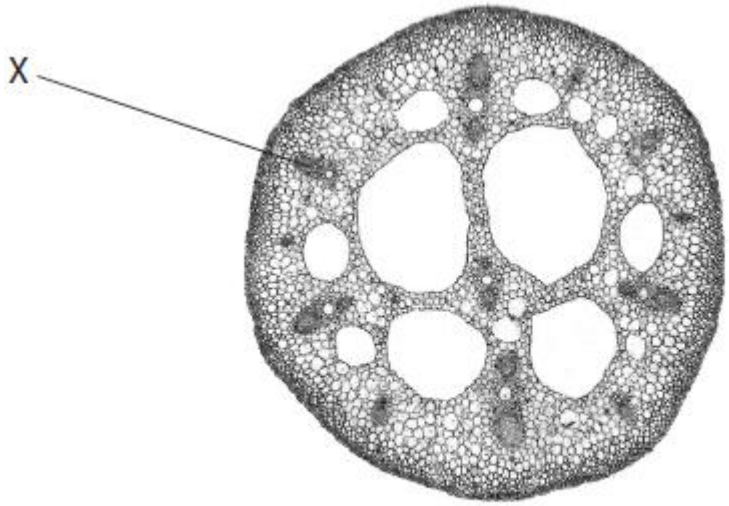
(1)

- ☒ **A** 7.3 m^2
- ☒ **B** 29.22 m^2
- ☒ **C** 730.6 cm^2
- ☒ **D** 2922.5 cm^2

(b) The photograph shows a cross-section of the stem of a water lily.

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(Source: © Scenics & Science/Alamy Stock Photo)

The structure labelled X contains xylem vessels and phloem.
How many of the following statements are correct?

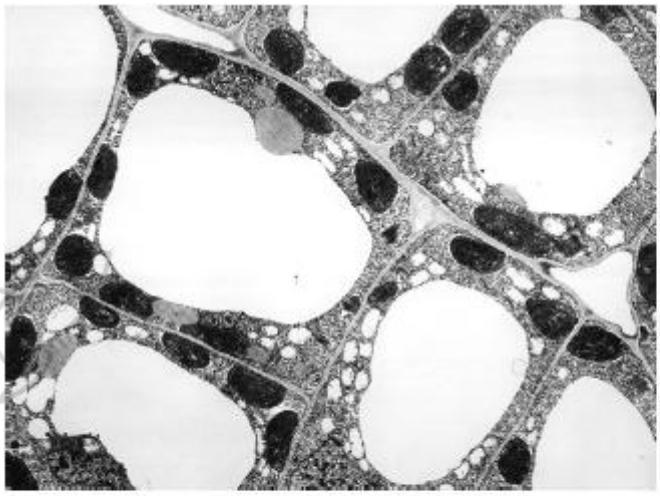
- both xylem vessels and phloem transport dissolved substances
- only phloem transport dissolved substances in one direction
- only the walls of xylem vessels contain both cellulose and lignin

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(c) The leaves of the water lilies contain several types of tissues and cells.

The photograph shows a group of leaf mesophyll cells, as seen using an electron microscope.



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Magnification × 7 800

(Source: © DR. JEREMY BURGESS/SCIENCE PHOTO LIBRARY)

(i) Label one permanent vacuole on this photograph.

(1)

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(ii) Give two functions of a vacuole.

(2)

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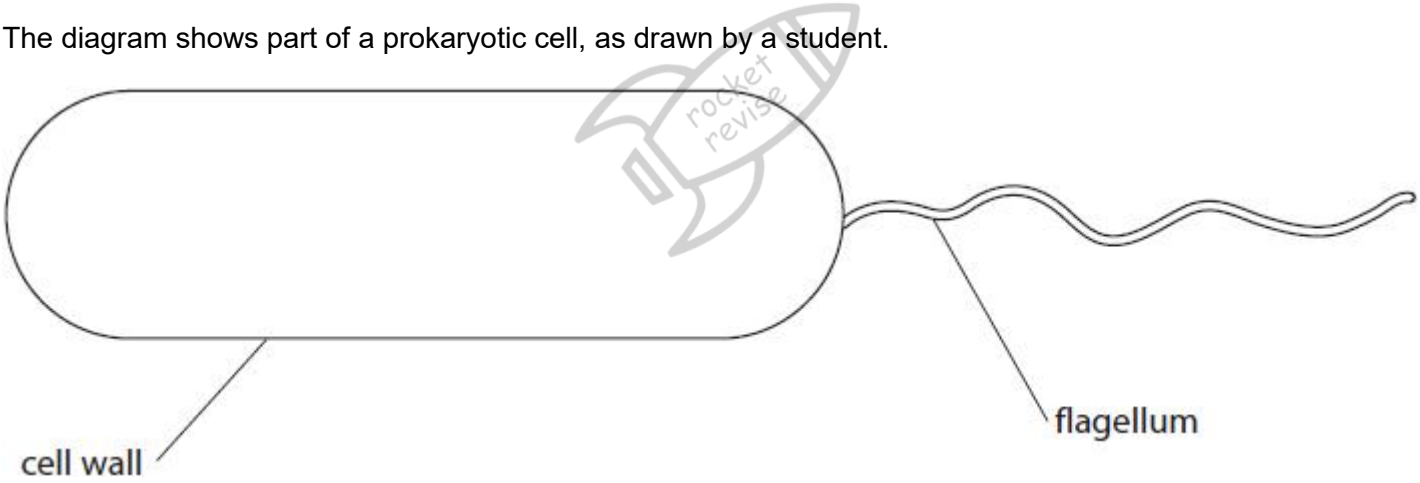
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(Total for question = 5 marks)

Q21.

The diagram shows part of a prokaryotic cell, as drawn by a student.



(a) Draw **and** label a capsule and two pili on the diagram.

(2)

(b) The effect of concentrations of sodium chloride solution on the growth of bacteria was investigated.

The same number of *L. piscium* bacteria were added to test tubes containing different concentrations of sodium chloride solution.

The volume of sodium chloride solution was the same in each test tube.

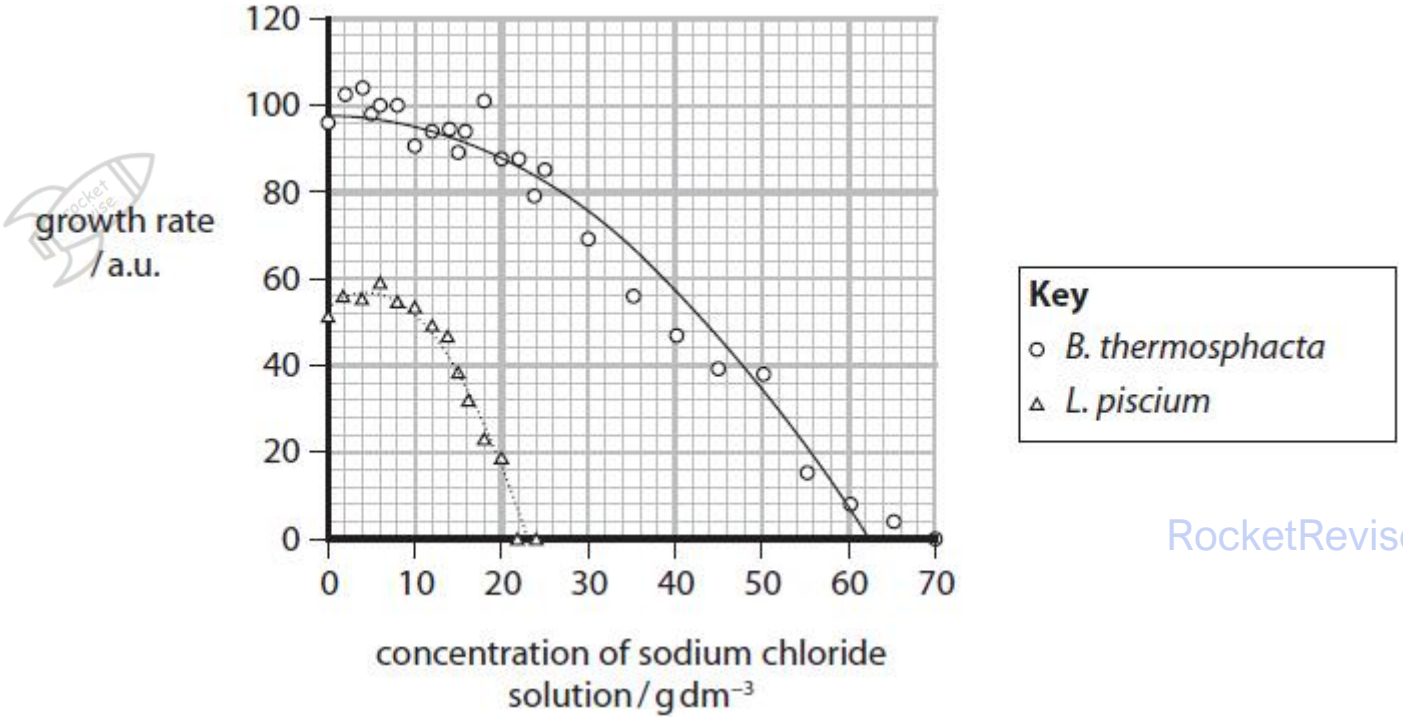
The growth rates of the bacteria were recorded after a set period of time.

The investigation was repeated with *B. thermosphacta* bacteria.

The graph shows the results of this investigation.

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Comment on the results of this investigation.

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(Total for question = 6 marks)

Q22.

Abaca plants are found in the Philippines.

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The photograph shows a mat made from abaca plant fibres.

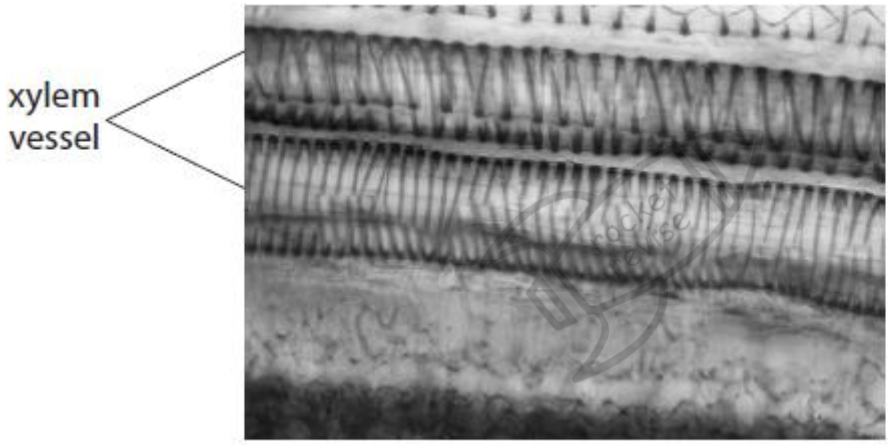


(Source: © Mehmet Çetin / Alamy Stock Photo)

Plant fibres include xylem vessels.

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(a) The photograph shows some xylem vessels, as seen using a light microscope.



(Source: © STEVE GSCHMEISSNER / SCIENCE PHOTO LIBRARY)

A stain was added to the xylem before viewing using a microscope.
Give a reason why it is usual to stain specimens in microscopy.

(1)

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(b) Explain how the arrangement of cellulose molecules and secondary thickening in xylem vessels contributes to the physical properties of the cell wall.

(4)

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(Total for question = 5 marks)

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Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a caterpillar and fluid inside a pitcher plant found on the island of Madagascar.

The fluid contains extracellular enzymes which are digesting the caterpillar.



(Source: © Martin Lindsay/Alamy Stock Photo)

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(a) Describe the role of the Golgi apparatus in the formation of these extracellular enzymes.

(2)

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(b) The digested insects are a source of nitrogen for the pitcher plant.

Give **two** ways in which this source of nitrogen is used by the pitcher plants.

(2)

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(c) Bacterial cells have also been found in the fluid in pitcher plants.

(i) Which **three** structures are found in both plant cells and bacterial cells?

(1)

- ☒ **A** cell membrane, cellulose cell wall and DNA
- ☒ **B** cell membrane, DNA and pili
- ☒ **C** cell wall, DNA and capsule
- ☒ **D** cell wall, DNA and ribosome

(ii) The fluid of the pitcher plant contains water and molecules from digested organisms, which bacterial cells need for growth.

Explain **two** other conditions needed for optimum bacterial growth in the fluid of the pitcher plant.

(2)

1 .....

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2 .....

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(Total for question = 7 marks)

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Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photographs show plants growing in vertical and horizontal systems.

In both of these systems, the plants grow in nutrient solutions instead of in soil.

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vertical system

(Source: © Lano Lan/Alamy Stock Photo)



horizontal system

(Source: © RossHelen editorial/Alamy Stock Photo)

(a) (i) The nutrient solutions contain magnesium ions and nitrates.

Describe the roles of these inorganic ions in the plants.

(2)

Magnesium ions .....

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Nitrates .....

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(ii) Suggest **one** advantage of the vertical system over the horizontal system.

(1)

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(b) Fruits contain cells.

The effect of calcium ion concentration on the firmness of fruits was investigated.
Cherry plants were treated once, twice or three times, with two different concentrations of calcium chloride solution. The firmness of the cherry fruits from these plants was then measured.
The table shows the effect on the mean fruit firmness of these treatments and a control.

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Number of treatments	Mean fruit firmness / N cm^{-2}	
	Plants treated with a calcium chloride solution of 950 mg dm^{-3}	Plants treated with a calcium chloride solution of 3 800 mg dm^{-3}
0 (control)	4.03	4.03
1	4.08	4.18
2	4.09	4.28
3	4.19	4.61

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(i) Which is the percentage increase in mean fruit firmness between the cherries from the control plants and those from plants treated three times with the higher calcium chloride concentration?

(1)

- ☒ **A** 3.72 %
- ☒ **B** 3.97 %
- ☒ **C** 6.20 %
- ☒ **D** 14.39 %

(ii) Give **one** conclusion for this investigation.

(1)

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(iii) Explain the effect of calcium ion concentration on the firmness of cherry fruits.

(3)

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(Total for question = 8 marks)

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Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a corpse lily flower (*Rafflesia hasseltii*).



(Source: © paul kennedy/Alamy Stock Photo)

The corpse lily plant is found in the rainforests of Sumatra.

The flowers can be 100 cm in diameter and are the largest flowers found on Earth.

(a) The smallest flower on Earth is produced by *Wolffia globosa*.

The flower of *W. globosa* measures 50 µm in diameter.
Which is the ratio for the diameters of the flowers of *R. hasseltii to W. globosa*?

(1)

- ☐ **A** 20 : 1
- ☐ **B** 200 : 1
- ☐ **C** 2 000 : 1
- ☐ **D** 20 000 : 1

(b) The corpse lily plant is unusual because none of its cells contains chloroplasts.

Most of the corpse lily plant grows inside the tissues of a rainforest vine plant, including inside the tissues involved in the translocation of organic solutes.
Only the corpse lily flower can be seen growing outside of the vine.
The corpse lily plant obtains organic solutes such as sucrose from one of the tissues in the vine plant.
(i) State what is meant by the term **tissue**.

(1)

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(ii) Suggest how the corpse lily plant obtains organic solutes, such as sucrose, from one of the tissues in the vine plant.

(3)

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(c) The wound healing properties of an extract made from corpse lily plants were investigated.

Treatments were tested on four groups of six rats with identical skin wounds.
The treatments were applied to the wounds twice a day.
The time taken for the wounds to heal was recorded.
The table shows the treatment applied to each group and the results of this investigation.

Rat group	Treatment	Mean healing time / days	Standard deviation
1	gel + no extract	21.67	±0.48
2	gel + 5% extract	15.83	±0.33
3	gel + 10% extract	14.67	±0.56
4	gel + current wound treatment	12.33	±0.31

(i) Explain the purpose of the gel used in these treatments.

(3)

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(ii) Comment on the results of this investigation.

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(Total for question = 12 marks)

Q26.

The use of plant products can contribute to sustainability.

(a) Fruits from calabash plants are used to make bowls.

The photograph shows a bowl made from a Nigerian calabash fruit.

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(Source: © Irene Abdou / Alamy Stock Photo)

(i) A spherical calabash fruit was cut in half to make two bowls.
Each bowl had a radius of 25 cm.
Calculate the volume of one bowl to the nearest whole number.

Use the formula: volume of a sphere = $\frac{4}{3}\pi r^3$

(2)

Answer ..... cm^3

(ii) Explain why bowls made from calabash fruits are a sustainable resource.

(2)

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(b) Fibres from the stems of the calabash plants have many uses. Some of these fibres are formed from phloem, sclerenchyma and xylem.

(i) How many of the following statements about sclerenchyma fibres are correct?

- they provide support to the plant
- they are used to translocate organic solutes
- they are used to transport water and mineral ions

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

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(ii) Compare and contrast the structures of phloem sieve tubes and xylem vessels.

(4)

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(Total for question = 9 marks)

Q27.

Boards used for chopping fruit and vegetables can be made from wood or oil-based plastics.

The photograph shows some chopping boards.



(Source: © Dorling Kindersley ltd/Alamy Stock Photo)

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(a) (i) Bacteria can grow on the surfaces of chopping boards.

Explain the conditions that would result in the greatest bacterial growth on the surface of a chopping board.

(3)

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(ii) Explain why boards made from oil-based plastics are not considered sustainable.

(2)

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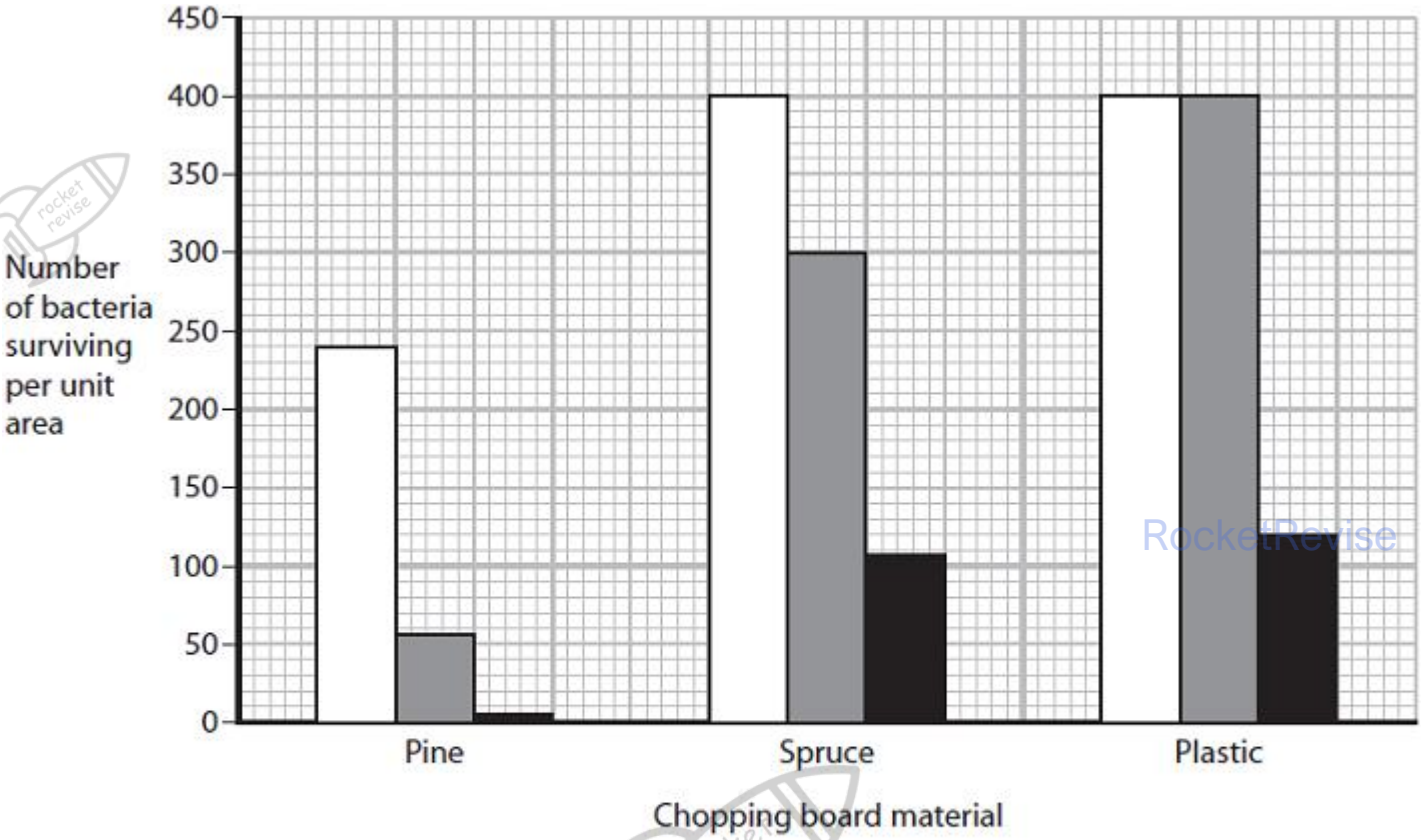
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(b) The number of bacteria surviving on boards made from different materials was investigated.

Identical bacterial cultures were placed on boards made of pine wood, spruce wood and plastic. The graph shows the number of bacteria surviving after bacteria were placed on the boards and kept in the same conditions.

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(i) Comment on the results of this investigation.

(3)

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(ii) Suggest **one** reason for the difference in bacterial survival on the pine chopping board and on the plastic chopping board.

(1)

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(Total for question = 9 marks)

Q28.

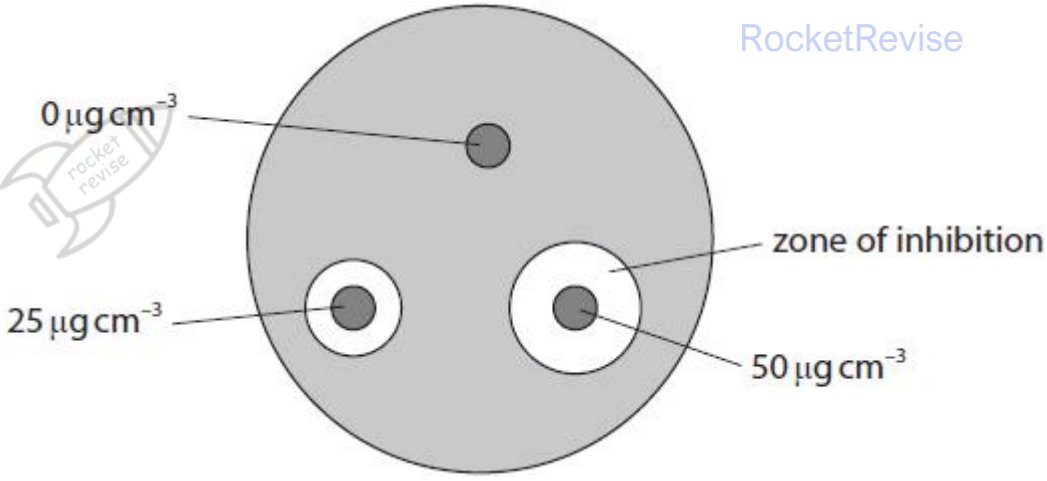
The photograph shows a Hawaiian parakeet flower (*Heliconia psittacorum*).



(Source: © H. Mark Weidman Photography/Alamy Stock Photo)

(a) The antimicrobial properties of an extract from this plant were investigated.

Three sets of agar plates were poured. Each set had a different type of bacteria spread over the surface of the agar.
Discs containing different concentrations of this extract were placed onto each plate, as shown in the diagram.
These plates were then incubated for 24 hours at 25 °C.



The diameter of each zone of inhibition was measured.
The table shows the results of this investigation.

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Extract concentration/ $\mu g\,cm^{-3}$	Mean diameter of zone of inhibition/mm		
	bacteria A	bacteria B	bacteria C
0	0.0	0.0	0.0
25	7.8	7.1	7.2
50	14.0	12.9	13.2

Comment on the antimicrobial properties of this extract.

(3)

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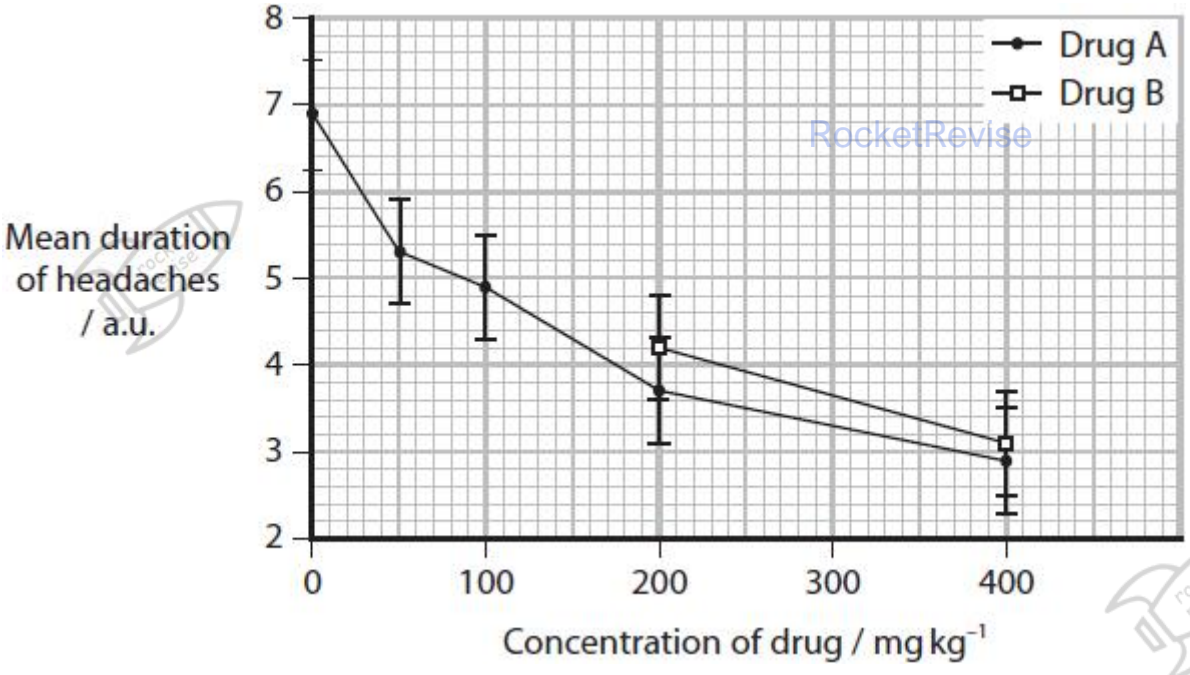
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(b) A new painkilling drug, drug A, has been developed from another species of plant.

This drug has been tested as a treatment for headaches.
The effect of this drug on the duration of headaches in volunteers was compared with drug B, a standard treatment for headaches.
The graph shows the results of this investigation.



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(i) Evaluate the results of this investigation.

(4)

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(ii) Drug A was then used in a three-phase drug trial.
Compare and contrast phases II and III in a contemporary drug trial using drug A.

(3)

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(Total for question = 10 marks)

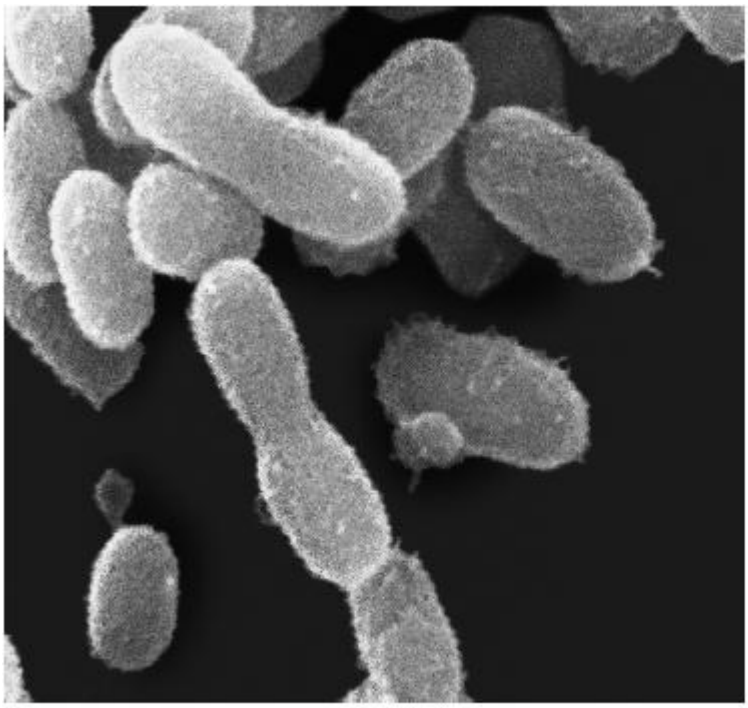
Q29.

A new type of bacteria was discovered in Greenland in 2008. It had survived for more than 120 000 years in the ice of a glacier.

The photograph shows this new type of bacteria, as seen using an electron microscope.

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(Source: © PENN STATE UNIVERSITY/SCIENCE PHOTO LIBRARY)

This type of bacteria can survive in low temperatures and low oxygen levels.

(a) Describe how scientists could determine that this was a new type of bacteria.

(2)

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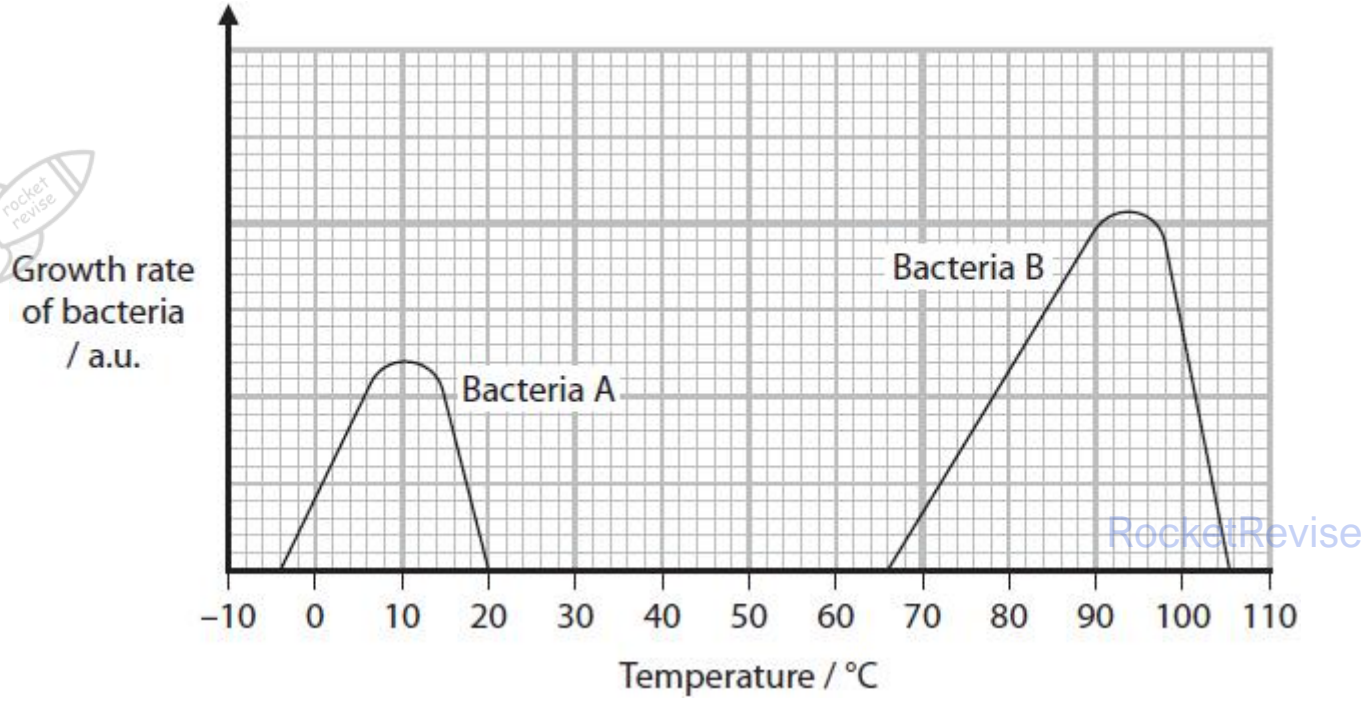
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*(b) Bacterial growth and survival can be affected by changing conditions.

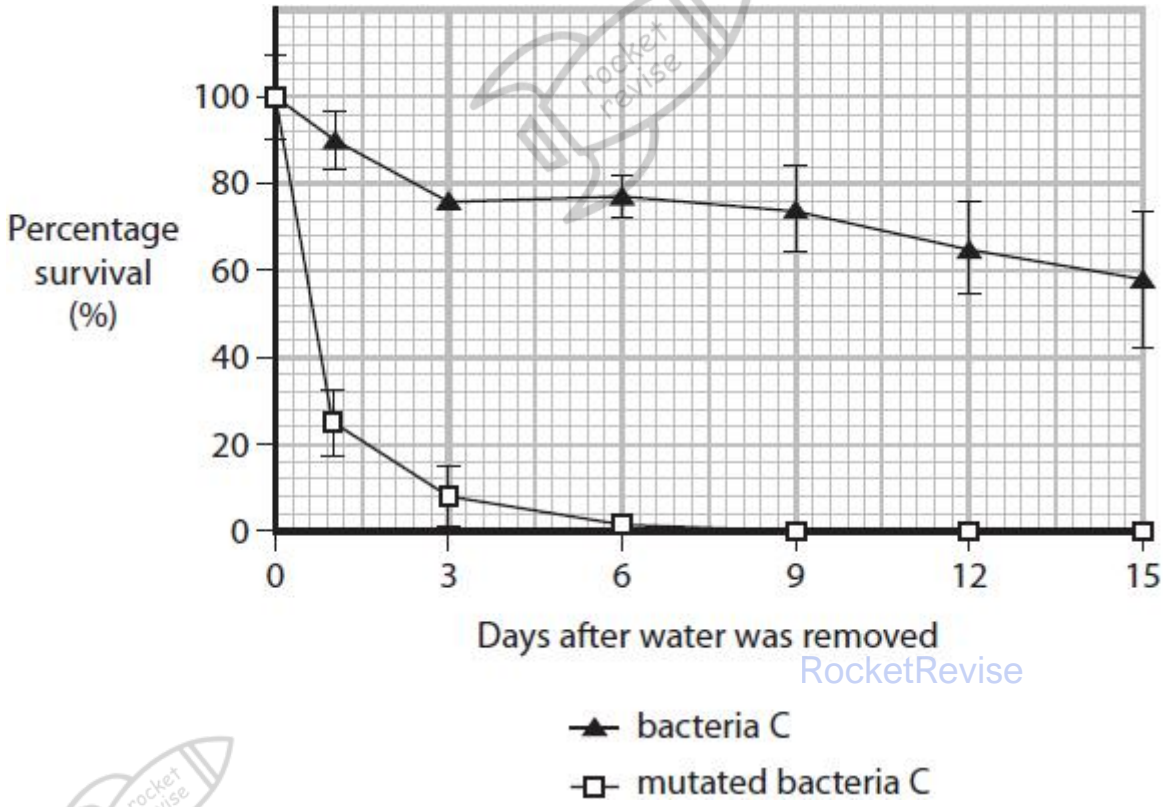
The effect of temperature and desiccation (water removal) on different bacteria was investigated. Graph 1 shows the effect of temperature on the growth rate of bacteria A and bacteria B.

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Graph 2 shows the percentage survival of bacteria C and mutated bacteria C after water was removed from the surroundings.



Explain the changes in growth rate and percentage survival of these bacteria.
Use the information in the graphs and your own knowledge to support your answer.

(6)

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(Total for question = 8 marks)

Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) Some plant cell walls develop secondary thickening.

Diagram 1 shows a cell without secondary thickening.
Diagram 2 shows part of the cell walls of two adjacent cells with secondary thickening.

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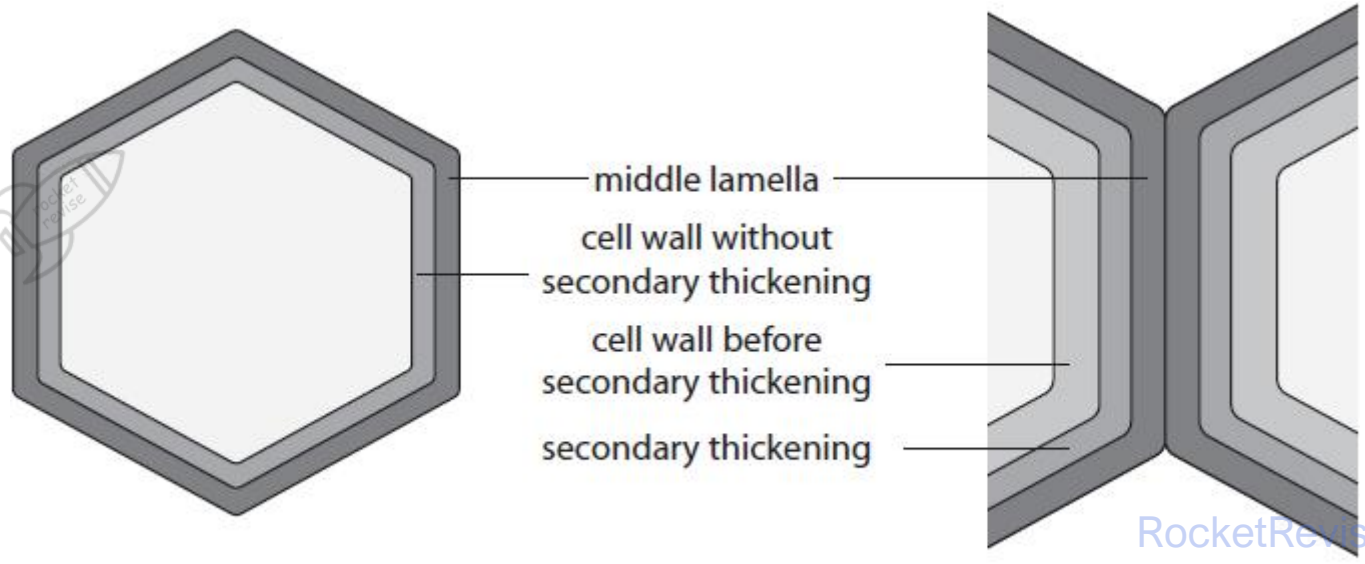


Diagram 1 Diagram 2

Give **three** differences between a cell wall with secondary thickening and a cell wall without secondary thickening.

Use the information in the diagrams to support your answer.

(3)

1 .....

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2 .....

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(b) (i) Plant cell walls contain cellulose.

How many of the following statements about cellulose molecules are correct?

- contain α-glucose molecules
- contain 1,4 and 1,6 glycosidic bonds
- form microfibrils

(1)

☒ **A** none
☒ **B** one
☒ **C** two
☒ **D** three

(ii) Secondary thickening occurs in sclerenchyma fibres.

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Explain how secondary thickening contributes to the physical properties of sclerenchyma fibres.

(2)

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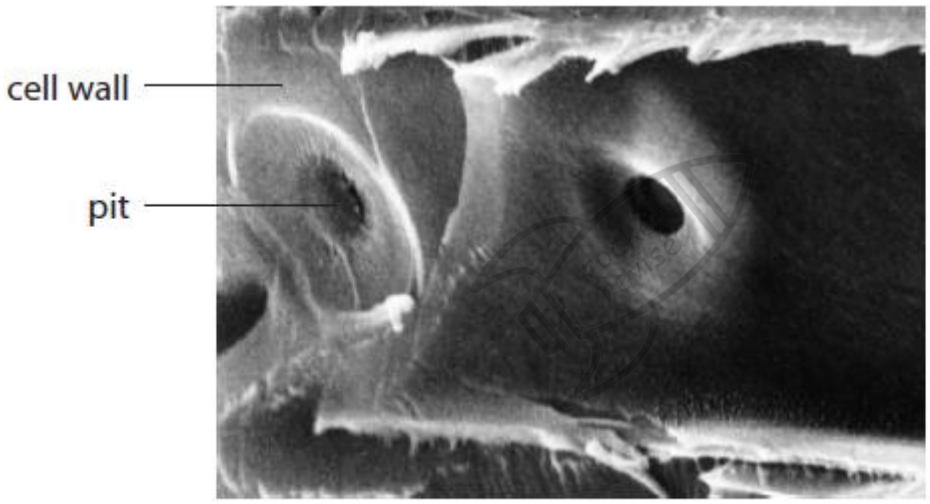
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(c) Some cells in xylem vessels have secondary thickening.

The photograph shows pits in the wall of part of a xylem vessel, as seen using an electron microscope.

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(Source: G.F. Gennaro / Science Photo Library)

Describe the function of pits in the xylem vessel.

(2)

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(Total for question = 8 marks)

Q31.

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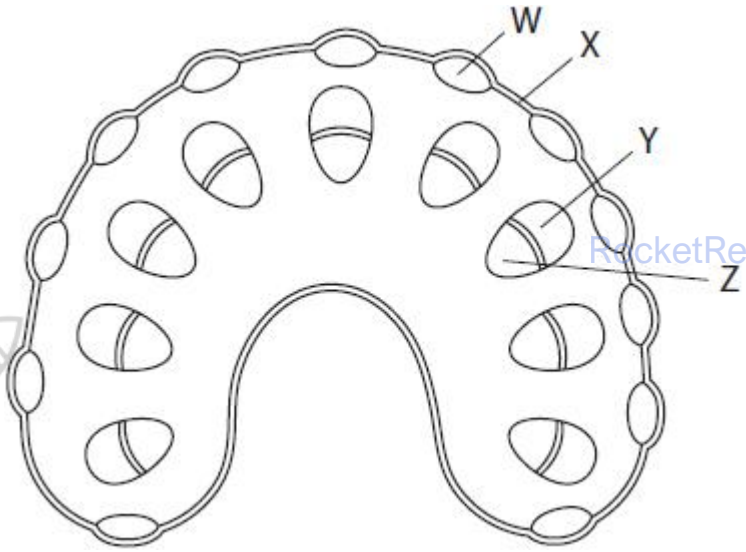
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows part of a celery plant which has been placed into a beaker of stain.



(Source: © Sara Sadler/Alamy Stock Photo)

A student made a drawing of a cross-section of the bottom of the stem of the celery plant before it was placed into the beaker of stain.



(a) One hour after being placed into the beaker of stain, the student observed a cross-section from half way up the celery stem.

(i) Which of the following would be a different colour due to the presence of the stain?

(1)

☒ **A** W

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☒ **B** X
☒ **C** Y
☒ **D** Z

(ii) Explain why, after one hour, this part of the stem would be a different colour.

(2)

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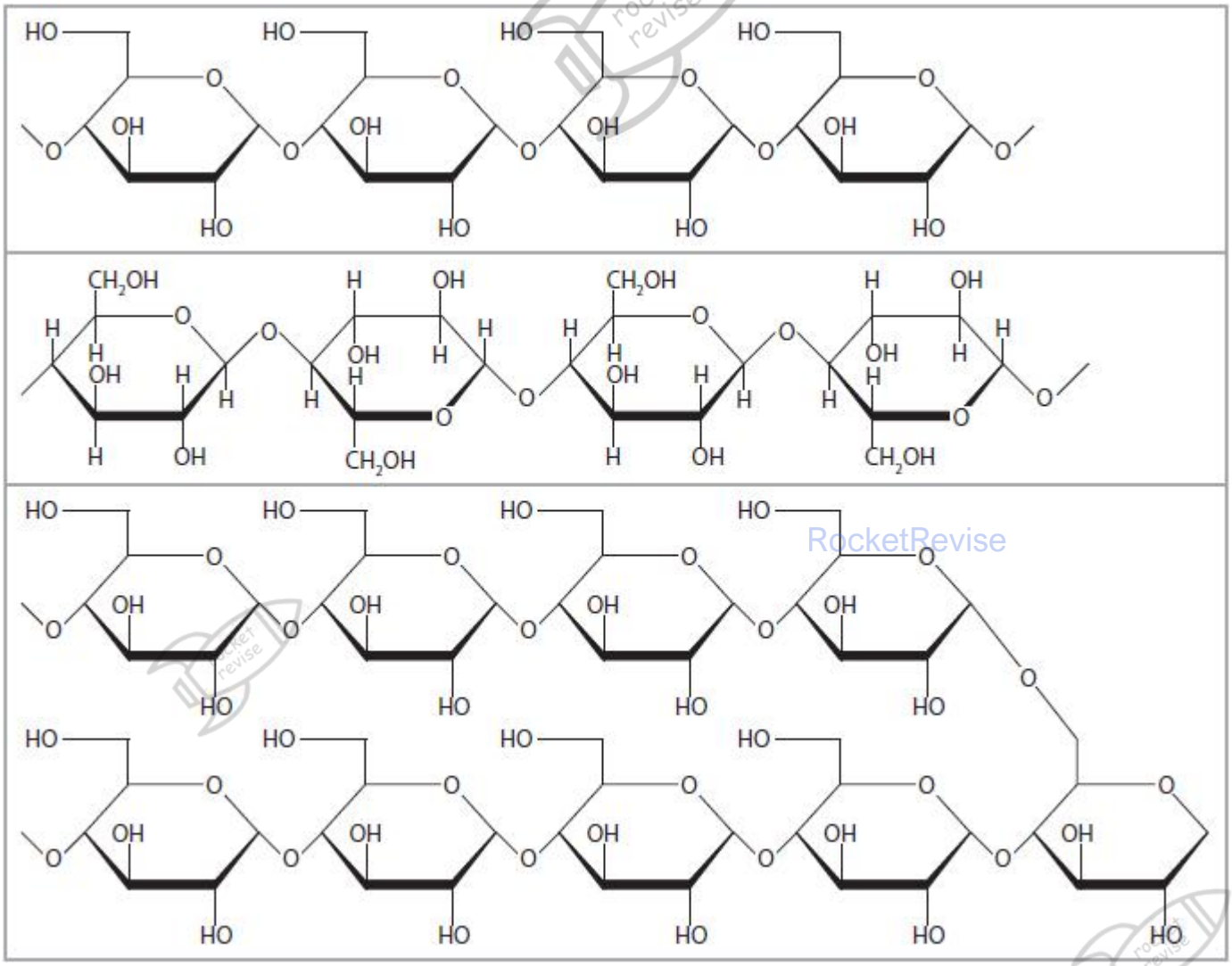
(b) Cells in the leaf contain organelles that store molecules of starch.

(i) Name **one** organelle in a leaf cell that stores starch.

(1)

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(ii) How many of the following show the structure of part of a starch molecule?



(1)

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- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(iii) Give **one** role of starch molecules in a leaf cell.

(1)

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(c) Transparent plastic film can be made from oil-based plastics or from starch.

The photograph shows transparent bio-plastic film, made from starch.

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(Source: © Roman/Alamy Stock Photo)

Give **one** reason why the use of starch-based transparent bio-plastic film is more sustainable than the use of oil-based plastics.

(1)

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(Total for question = 7 marks)

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Q32.

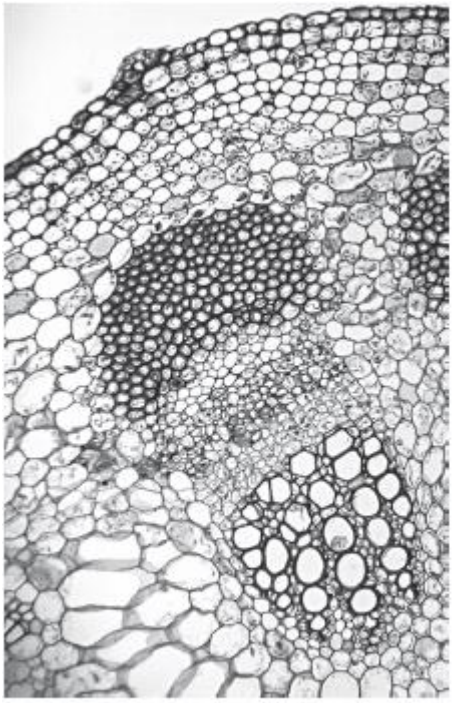
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Plant stems contain phloem, sclerenchyma fibres and xylem vessels.

(a) The photograph shows part of a transverse section through the stem of a plant.

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(Source: © Ed Reschke/Getty Images)

(i) Label the photograph to show the position of the phloem, sclerenchyma fibres and xylem vessels.

(3)

(ii) How many of the following statements about phloem are correct?

- connected to other cells by plasmodesmata
- transports water and organic solutes
- translocation moves substances in phloem

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(b) The cells of plants and prokaryotes have cell walls.

(i) Name a structural molecule in each type of cell wall.

(2)

Plant cell wall

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Prokaryotic cell wall

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(ii) Explain how the arrangement of molecules contributes to the physical properties of the cell walls of sclerenchyma fibres.

(3)

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(Total for question = 9 marks)

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Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The lack of some inorganic ions in soil can affect the yield of fruit from crop plants.

(a) The photograph shows part of a vine plant growing in soil lacking magnesium ions.

The vine plant shown has yellow leaves and reduced growth.



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(Source: Nigel Cattlin / Alamy Stock Photo)

(i) Which tissue is involved in the transport of magnesium ions in a plant?

(1)

- ☒ **A** amyloplast
- ☒ **B** plasmodesmata
- ☒ **C** sclerenchyma
- ☒ **D** xylem

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(ii) Explain how a lack of magnesium ions could result in yellow leaves and reduced growth.

(3)

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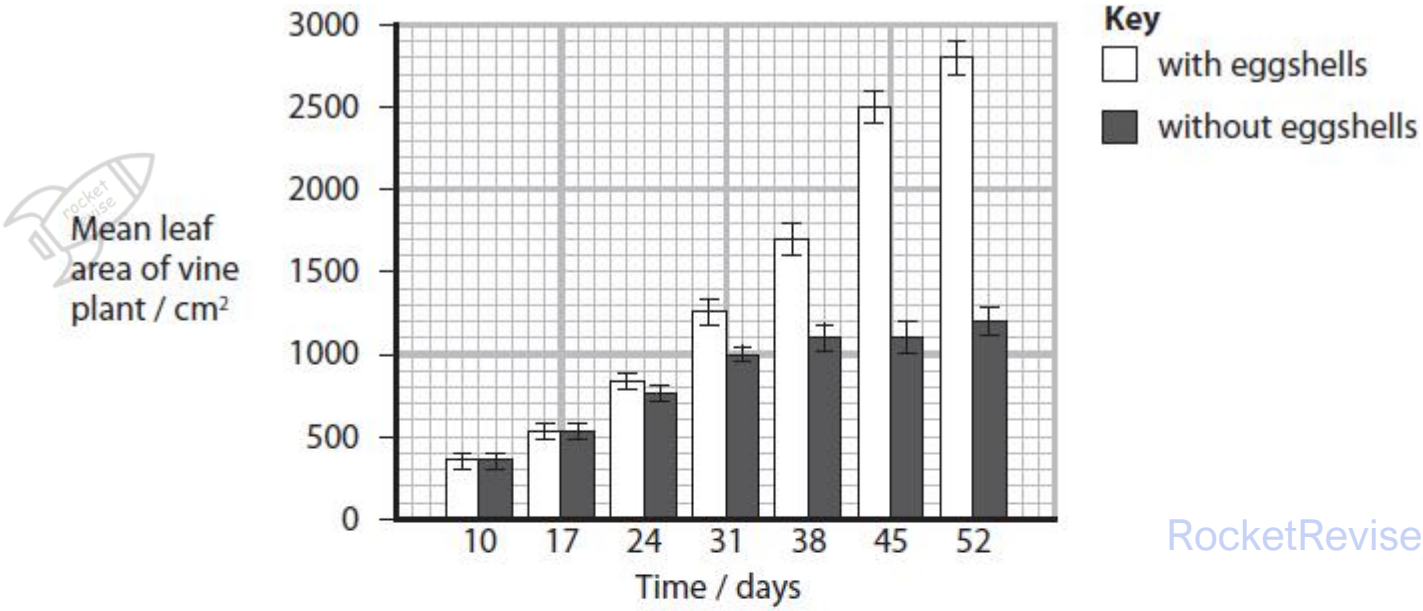
*(b) Calcium ions are also needed for the growth of vine plants and fruit production.

Chicken eggshells contain calcium ions. Some farmers add crushed eggshells to the soil surrounding vine plants.
The effect of adding crushed eggshells to the soil surrounding vine plants was investigated.
The mean leaf area, mean shoot length and mean chlorophyll content were calculated over 52 days.
The graphs show the results of this investigation.

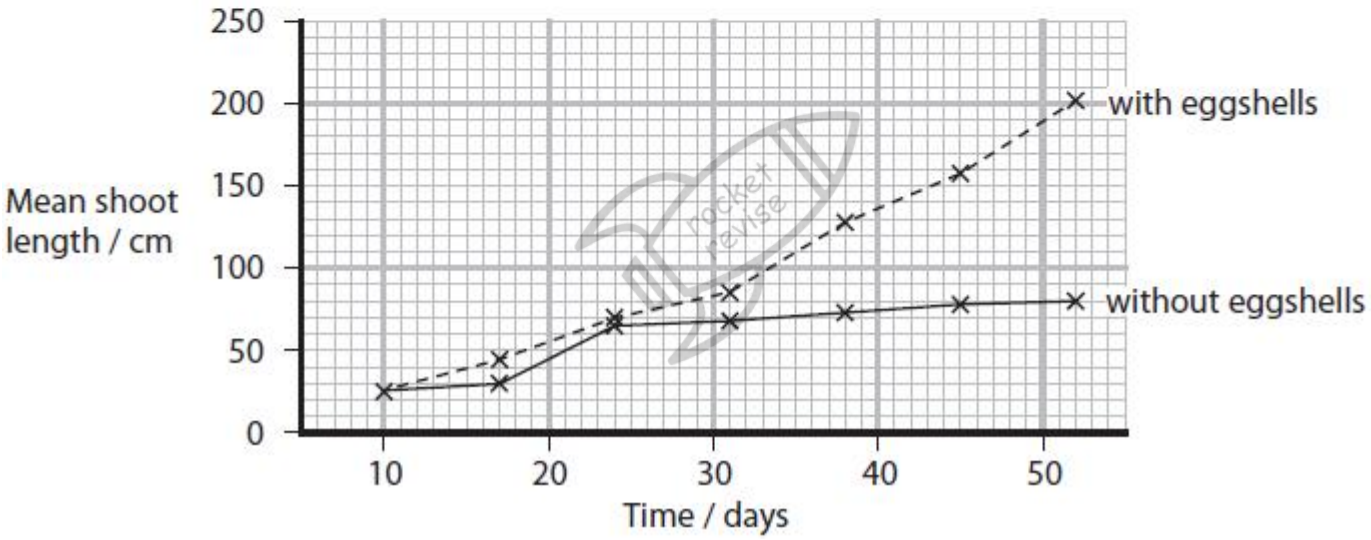
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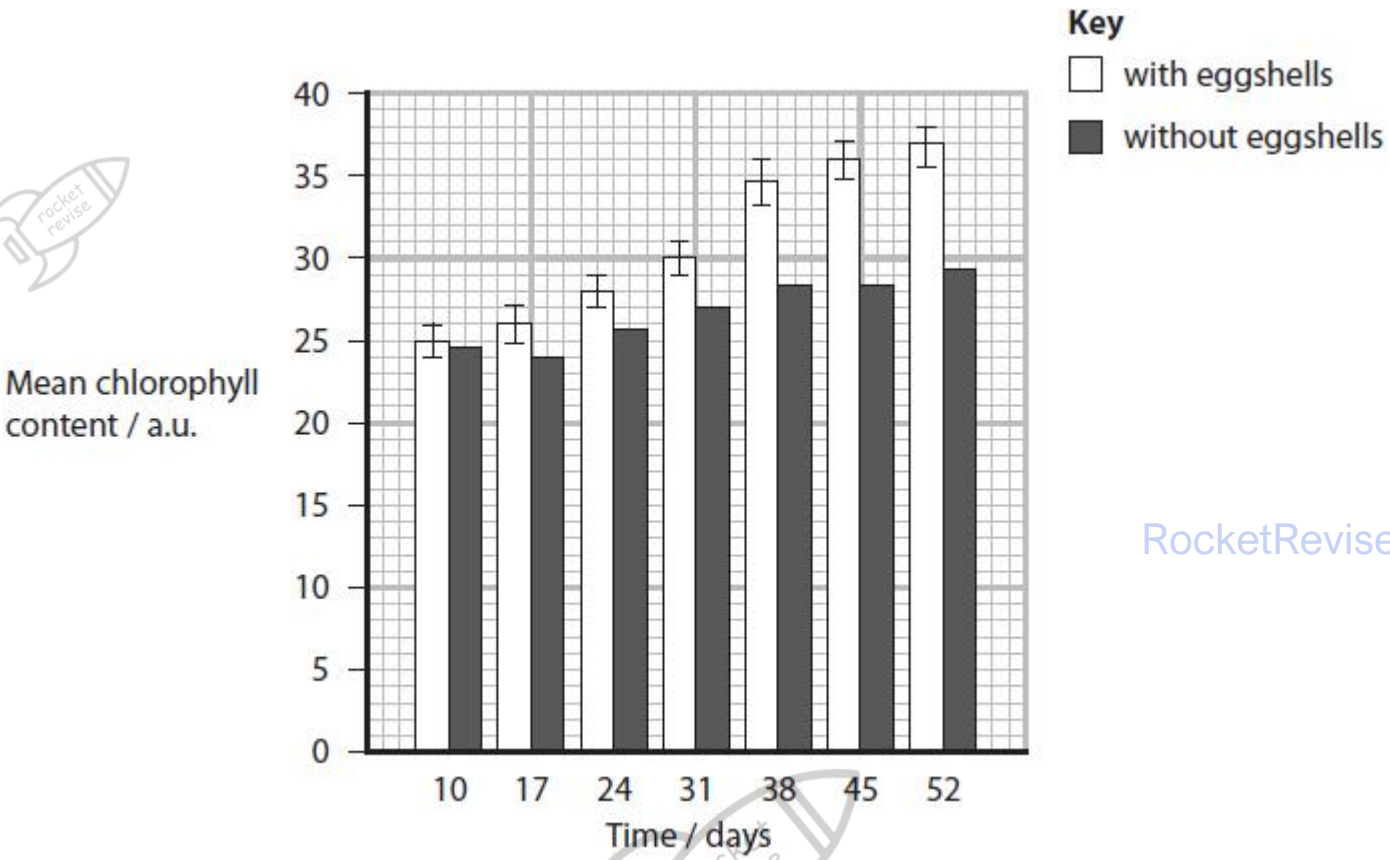
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Discuss the benefits of adding crushed eggshells to the soil surrounding vine plants.

Use the information in the question to support your answer.

(6)

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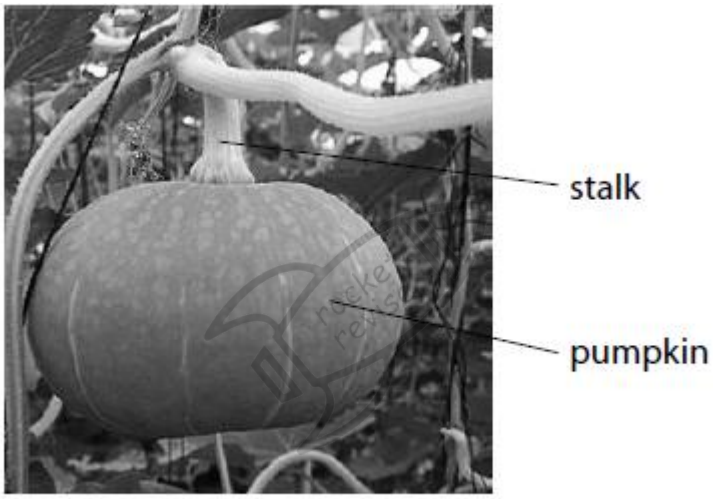
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(Total for question = 10 marks)

Q34.

The photograph shows a pumpkin.

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Source from: https://www.aliexpress.com/price/winter-outdoor-plants_price.html

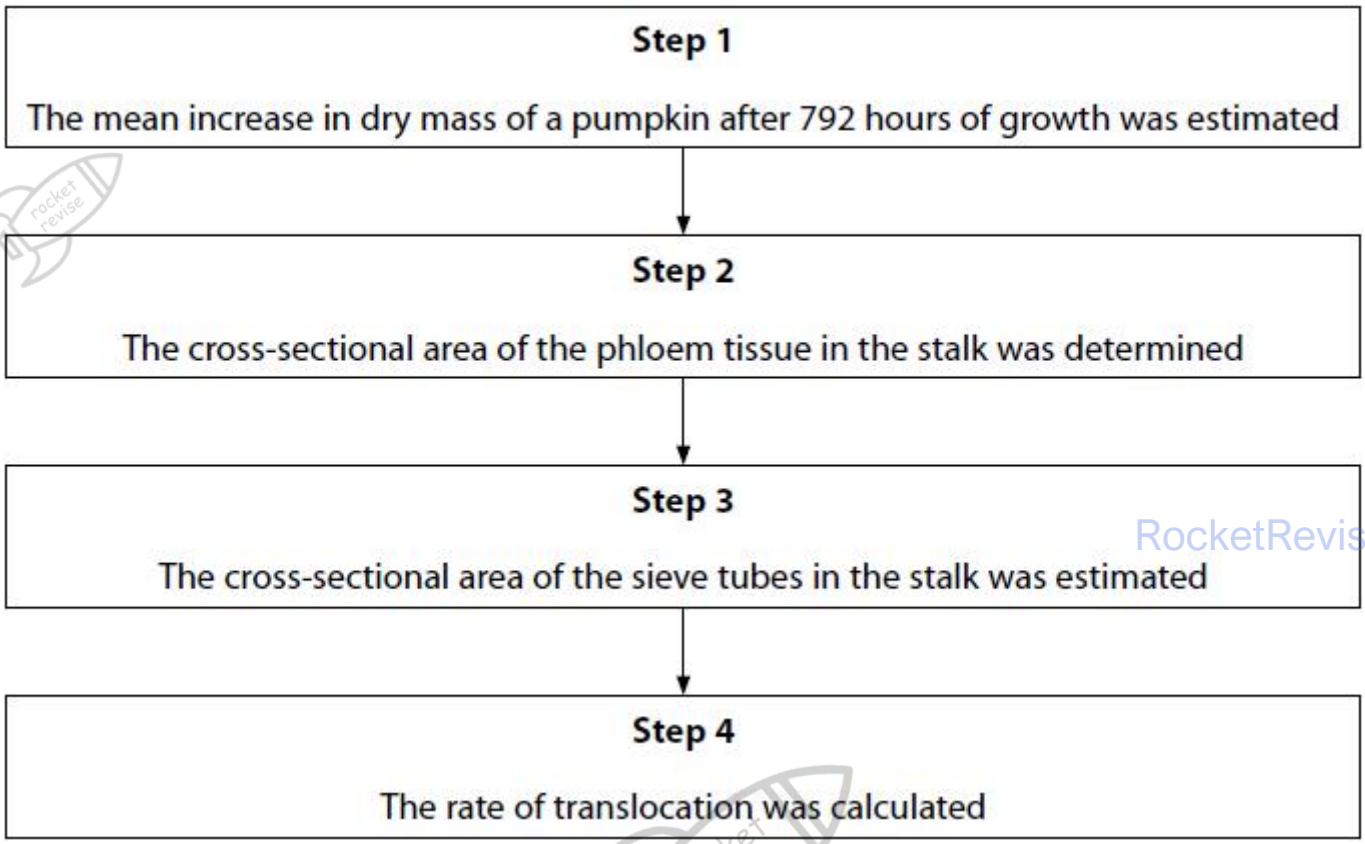
The scientists Crafts and Lorenz investigated the rate of translocation through the phloem in pumpkins.

The flow chart shows the method used in this investigation.

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(a) (i) Suggest how the mean increase in dry mass of a pumpkin could be estimated in **Step 1**.

(3)

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(ii) Explain why Crafts and Lorenz used dry mass in this investigation.

(2)

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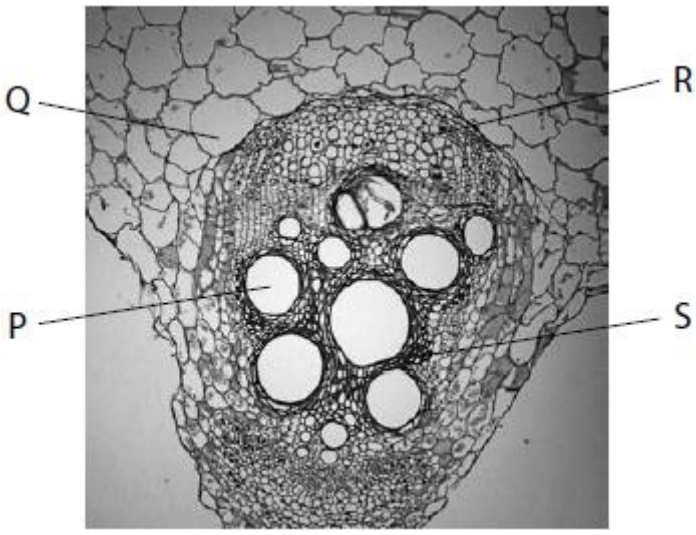
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(b) (i) The photograph shows a cross-section through part of a stalk, as seen using a light microscope.



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Which letter is pointing to the phloem?

(1)

- ☒ **A** P
- ☒ **B** Q
- ☒ **C** R
- ☒ **D** S

(ii) Describe a method that could be used to determine the cross-sectional area of the phloem in **Step 2**.

(2)

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(c) Give a reason why only the cross-sectional area of the sieve tubes, rather than the phloem tissue, was estimated in **Step 3**.

(1)

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(d) What are the units for the rate of translocation calculated in **Step 4**?

(1)

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- ☒ **A** g cm^{-2} hr^{-1}
- ☒ **B** g cm^{2} hr^{-1}
- ☒ **C** g cm^{-3} hr^{-1}
- ☒ **D** g cm^{3} hr^{-1}

(Total for question = 10 marks)

Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Fish are often packaged for transport to markets.

The bacteria *E.coli* and *S. aureus* are found on the surface of fish. These bacteria can cause disease in humans.

(a) Which structures enable *E.coli* to stick to the surface of fish?

(1)

- ☒ **A** capsule and flagellum
- ☒ **B** capsule and pili
- ☒ **C** flagellum and pili
- ☒ **D** flagellum and ribosomes

(b) High–density polyethylene (HDPE) and polystyrene are two examples of oil–based plastics.

The photograph shows polystyrene boxes that have been used to transport fish to a fish market.



(Source: © Roberto Fumagalli / Alamy Stock)

(i) Plant–based packaging is more sustainable than oil–based packaging.
Explain what is meant by **sustainable**.

(2)

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(ii) The effect of packaging material on the growth of viable bacteria on the surface of fish was investigated.

The number of *S. aureus* bacteria on four groups of fish was determined before each group of fish was placed in a different type of container.

The containers of fish were kept at 1 °C for one week.

The table shows the means and standard deviations for the decrease in the number of colonies of *S. aureus* bacteria.

Container packaging material	Mean decrease in the number of *S. aureus* bacteria / colonies cm^{-2}
Poplar wood	0.79 ± 0.44
Pine wood	7.14 ± 0.28
HDPE	0.00 ± 0.02
Polystyrene	0.08 ± 0.02

Comment on the results of this investigation.

(4)

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(Total for question = 7 marks)

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Q36.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

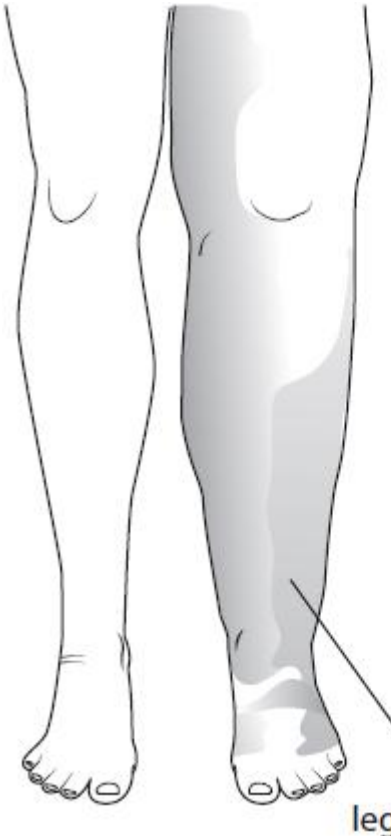
Cellulitis is an infection of the skin.

It is caused when bacteria living on the skin surface enter through a cut in the skin.

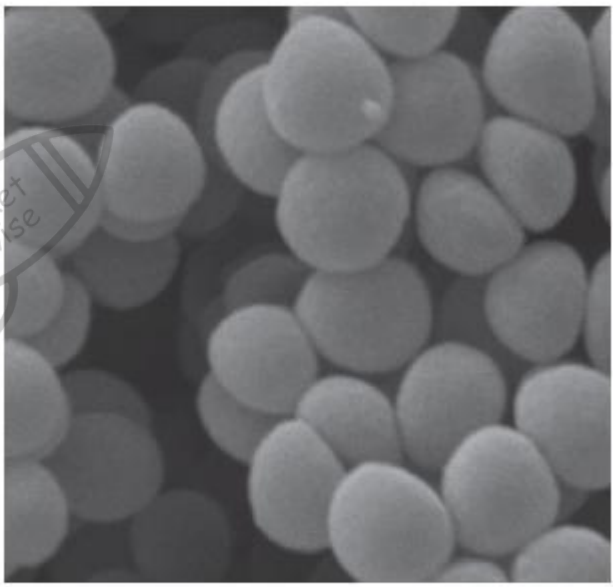
This results in swelling and redness of the infected area.

The images show a patient with cellulitis in one leg and one of the types of skin bacteria that cause cellulitis.

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leg with cellulitis



(Source: © Louis Lau / Alamy Stock Photo)

bacteria

(a) This type of bacteria has a capsule, but not a flagellum.

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(i) Suggest **one** advantage of having a capsule to this type of bacteria.

(1)

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(ii) Suggest why this type of bacteria does **not** need a flagellum.

(1)

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(iii) Which of the following structures would be found in a bacterial cell?
1. cell membrane
2. cell wall
3. ribosomes

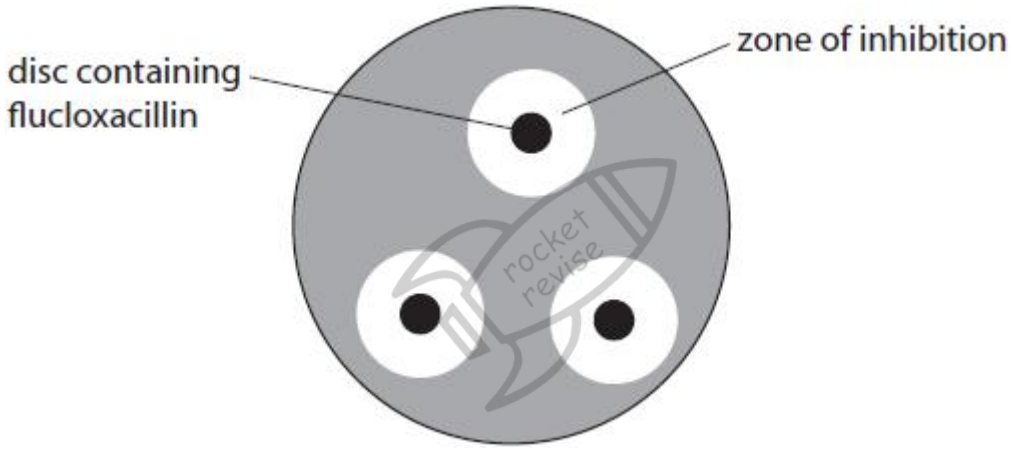
(1)

- ☐ **A** 1 and 2 only
- ☐ **B** 1 and 3 only
- ☐ **C** 2 and 3 only
- ☐ **D** 1, 2 and 3

(b) The antibiotic flucloxacillin is the current treatment for cellulitis.

A student investigated the antimicrobial properties of flucloxacillin.
An agar plate had the type of bacteria which causes cellulitis spread over the surface of the agar.
Three discs containing the **same** concentration of flucloxacillin were placed onto the plate, as shown in the diagram.
The plates were incubated for 24 hours at 25°C.

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The table shows the student's results.

Diameter of zone of inhibition / mm
11.0
7.5
8.5
Mean ($\bar{x}$) = 9.0

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Standard deviation can be calculated by using the formula:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

The table shows some of the calculations done by the student.

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x	$x - \bar{x}$	$(x - \bar{x})^2$
11.0	2.0	4.0
7.5		
8.5	−0.5	0.25
		$\sum(x - \bar{x})^2 =$

(i) Calculate the standard deviation for this set of data.
Use the student's table and formula to help you.

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(3)

Answer ..... mm

(ii) Which of the following explains why the student would want to calculate the standard deviation(s) (SD) for this experiment?

(1)

- ☒ **A** to see how widely the diameter values differ from the mean diameter
- ☒ **B** to see if the observed results differ from the expected results
- ☒ **C** to see if there is a correlation between different concentrations and zone of inhibition diameter
- ☒ **D** to see which is the most frequently occurring number

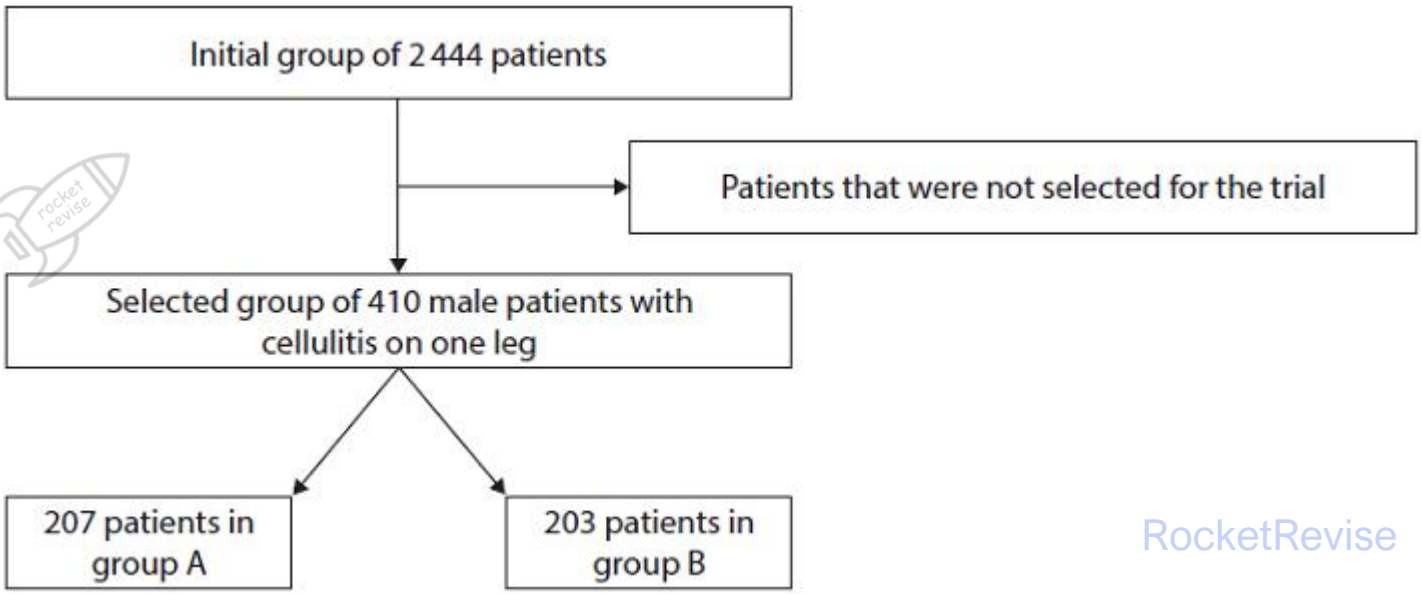
(c) The effectiveness of treating cellulitis with flucloxacillin alone was compared with a treatment that used both flucloxacillin and clindamycin.

The diagram shows how two groups of patients with cellulitis were selected from an initial group of 2 444 patients.

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(i) Which percentage of the initial group of patients were placed in **group B**?

(1)

- ☒ **A** 8.3 %
- ☒ **B** 8.5 %
- ☒ **C** 49.5 %
- ☒ **D** 50.5 %

(ii) Suggest how this study could be carried out as a double-blind trial.
Use the diagram to support your answer.

(1)

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(iii) The table shows the mean difference in circumference for the infected and non-infected legs at 5 and 10 days after treatment began.

Group	Mean difference in circumference between infected leg and non-infected leg / cm	
	After 5 days	After 10 days
A	2.18	1.77
B	2.04	1.42

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Comment on the results of this investigation.

(4)

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(Total for question = 13 marks)

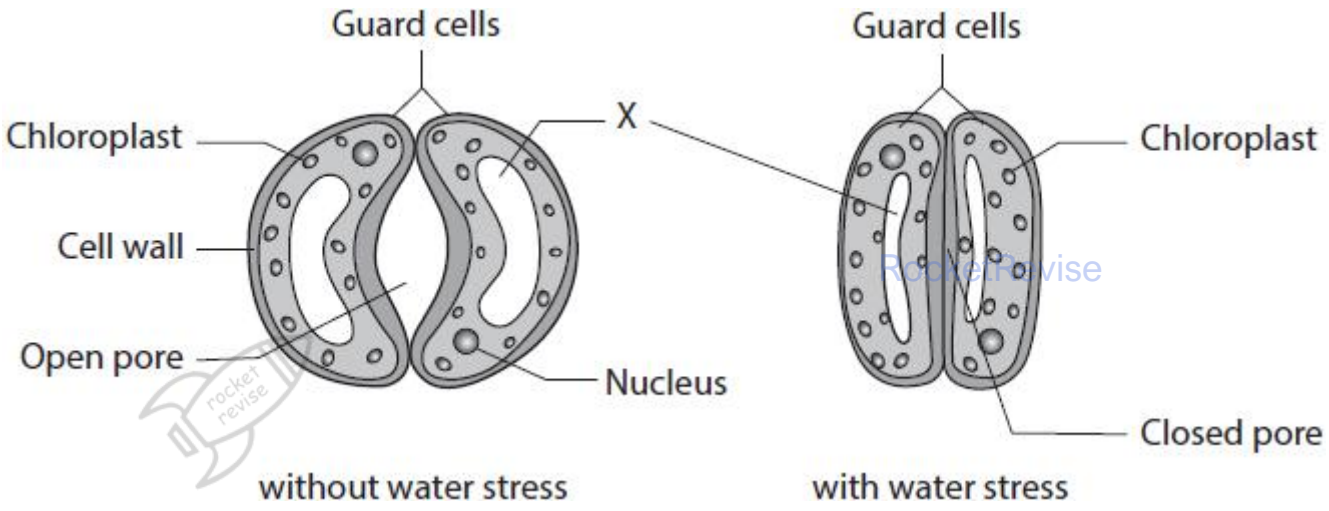
Q37.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Water is important for plants.

There are pores in the surface of the leaf, which are formed by guard cells. Changes to the guard cells cause the pores to open or close.

The diagram shows guard cells, at the same time of day, from a wheat plant without water stress and from a wheat plant that has water stress.



When a plant is not given sufficient water, for a period of time, it develops water stress.

This can have several effects on the plant.

(a) One effect is to close the pores in the leaves. This occurs due to changes in the guard cells.

(i) Name the structure labelled X.

(1)

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(ii) Suggest why the closing of the pores reduces water stress in the plant.

(1)

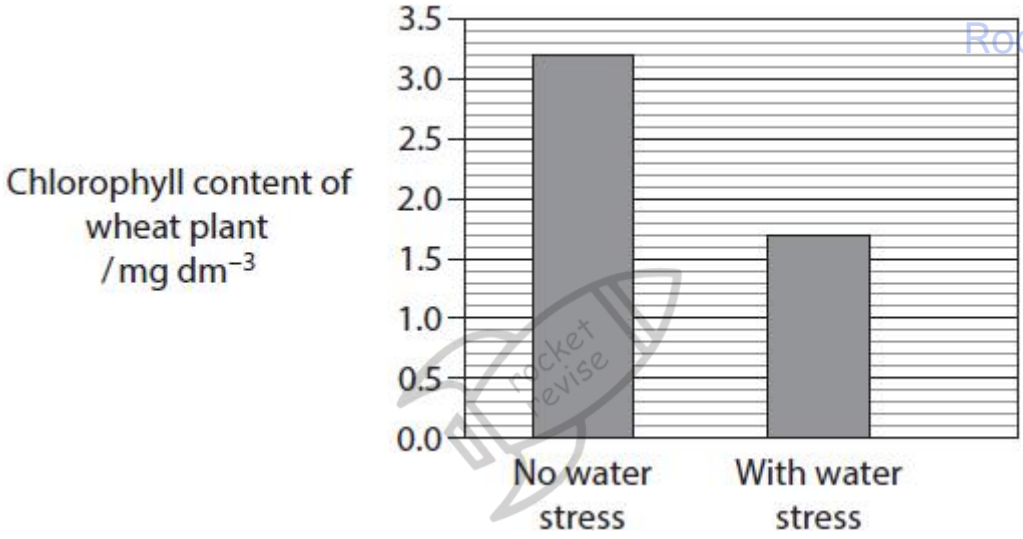
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(b) Another effect is the reduction in the absorption of mineral ions, for example magnesium ions.

Magnesium ions are needed in the formation of chlorophyll.
The effect of water stress on the chlorophyll content of a wheat plant was investigated.
The results are shown in the graph.



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Which is the percentage change in chlorophyll content when the wheat plant has water stress?

(1)

- ☒ **A** +46.9 %
- ☒ **B** −46.9 %
- ☒ **C** −88.2 %
- ☒ **D** +88.2 %

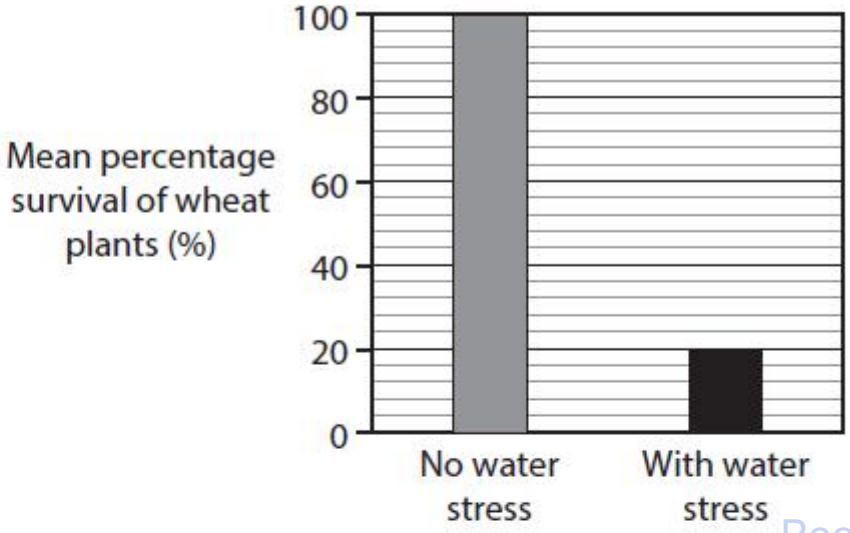
*(c) The graphs show three more effects of water stress on wheat plants.

Graph A shows the effect of water stress on percentage survival rate.

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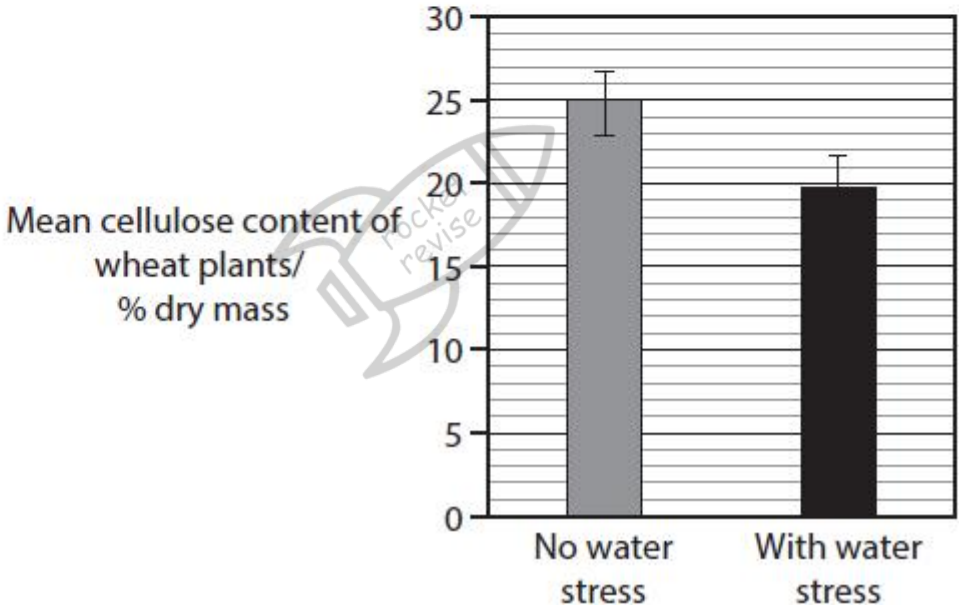
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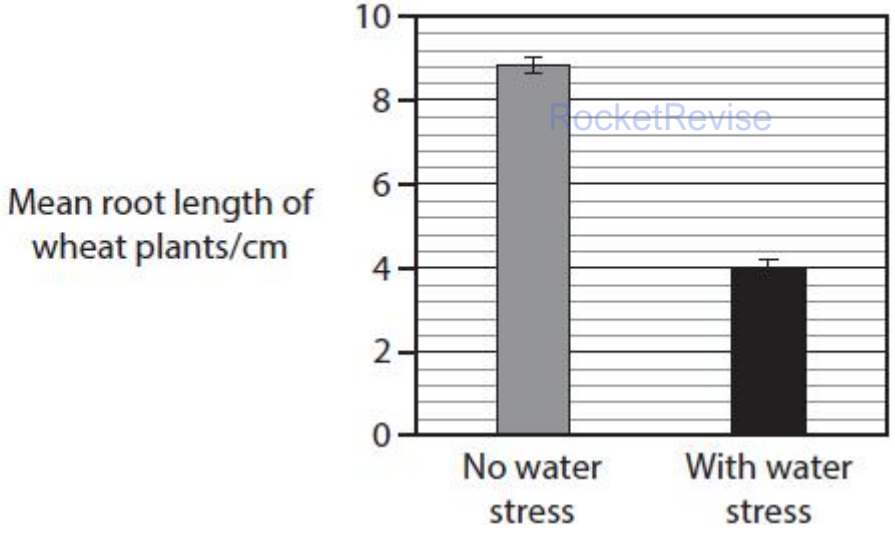


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Graph B shows the effect of water stress on cellulose content.



Graph C shows the effect of water stress on root length.



Deduce how water stress could affect the growth of wheat plants.

Use information from the whole of the question and your own knowledge to support your answer.

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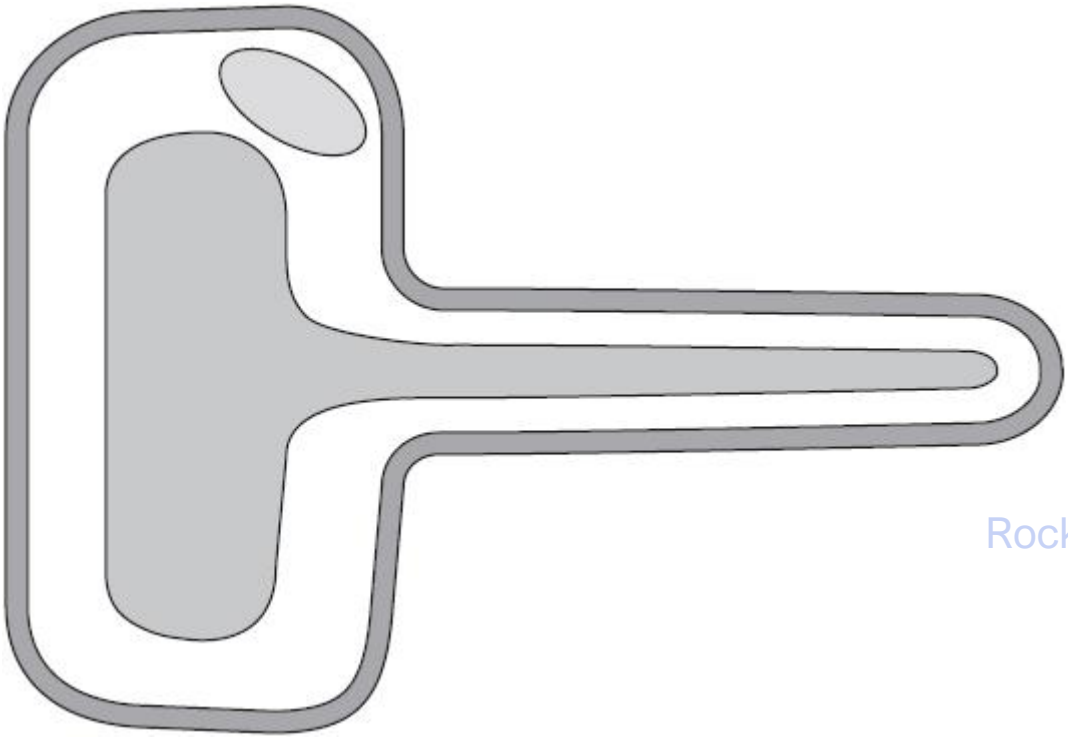
(Total for question = 9 marks)

Q38.

The diagram shows an incomplete plant cell, as drawn by a student.

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(a) (i) Label the **tonoplast** on the diagram.

(1)

(ii) Draw **and** label a **nucleolus** in this plant cell.

(1)

(b) Complete the table to show which of these structures would be present or absent in plant leaf cells.

(2)

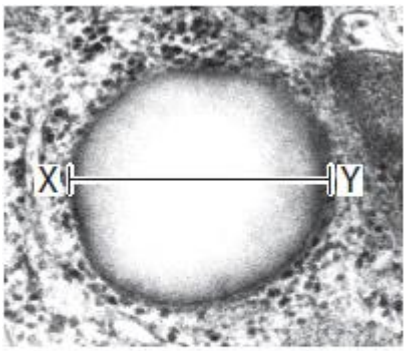
Structure	✓ if present × if absent
amyloplast	
chloroplast	
Golgi apparatus	
plasmodesmata	

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(c) The photograph shows a lysosome, as seen using an electron microscope.

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(Source: © Science History Images / Alamy Stock Photo)

Magnification ×250 000

Calculate the actual width (X–Y) of this lysosome in μm.

Give your answer to **two** decimal places.

(1)

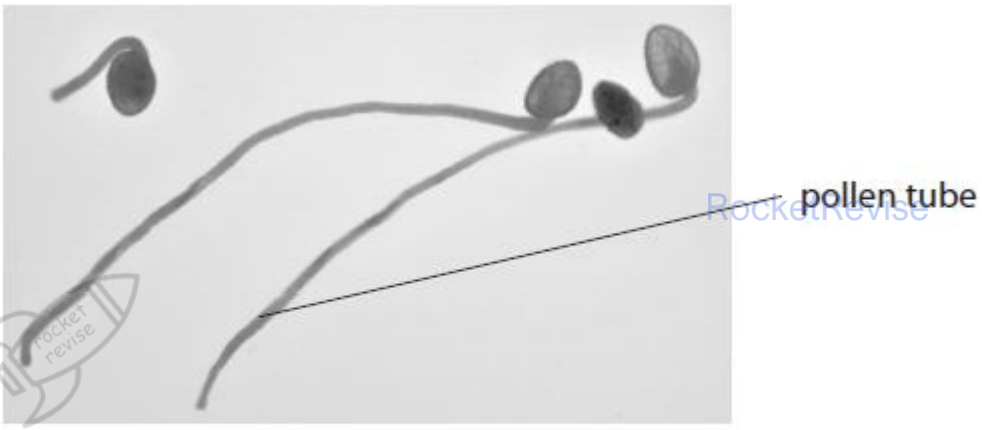
Answer ..... μm

(Total for question = 5 marks)

Q39.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows some pollen tubes growing from pollen taken from a lily flower, as seen using a light microscope.



Magnification ×100

(Source: © agefotostock/Alamy Stock Photo)

(a) (i) The pollen tube growth rate is 0.2 μm s^{-1}.

How long would it take the pollen tube to grow 15 mm?

(1)

☒ **A** 75 seconds

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- **B** 125 minutes
- **C** 1 250 minutes
- **D** 7 500 seconds

(ii) State how to calculate the actual length of one of the pollen tubes in the photograph.

(1)

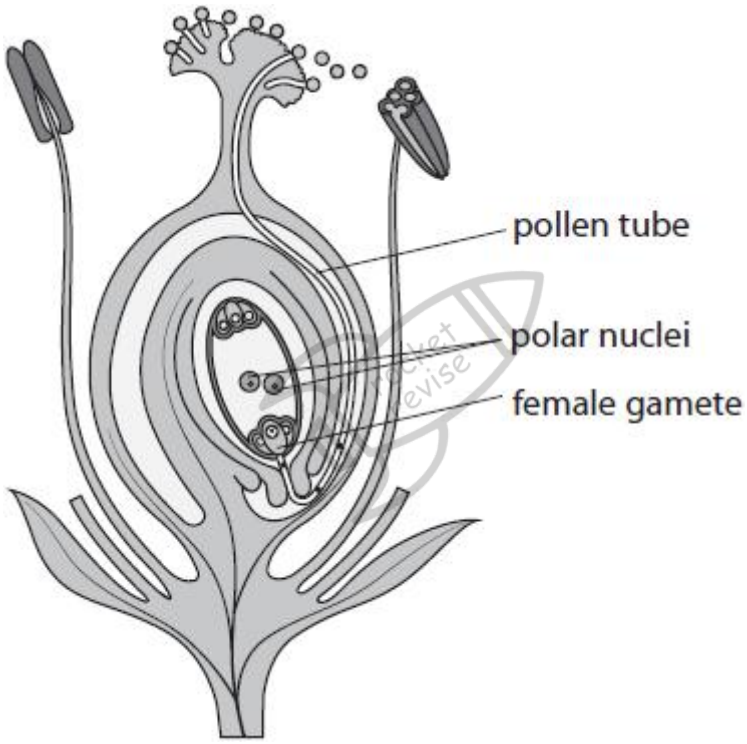
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(b) The diagram shows a pollen tube that has grown to reach an ovule.

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Fertilisation of the female polar nuclei results in the production of endosperm tissue.
Fertilisation of the female gamete results in the production of an embryo.
Explain why an embryo cell and an endosperm cell contain **different** quantities of DNA.

(4)

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(c) The photograph shows part of a Madonna lily plant (*L. candidum*).



(Source: © Zoonar GmbH/Alamy Stock Photo)

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An extract made from *L. candidum* has been used in traditional medicine as a treatment for human skin burns.
Describe a drug trial method that a scientist could use to determine if this extract is:

- safe to use in humans
- more effective than the current skin burn treatment used in a hospital.

(5)

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(Total for question = 11 marks)

Q40.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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All living organisms are made of cells.

(a) The table shows some structures found in living cells.

Complete the table to show in which type of cells these structures are present (✓) or absent (✗).

(4)

Structure	Animal cell ✓ = present ✗ = absent	Plant cell ✓ = present ✗ = absent	Prokaryotic cell ✓ = present ✗ = absent
amyloplast			
circular DNA			
mitochondria			
nucleolus			

(b) The smallest living cell has a diameter of 0.2 μm.

The volume of a sphere is calculated using the formula:

$$\frac{4}{3}\pi r^3$$

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Which of the following shows the volume of this cell in standard form?

(1)

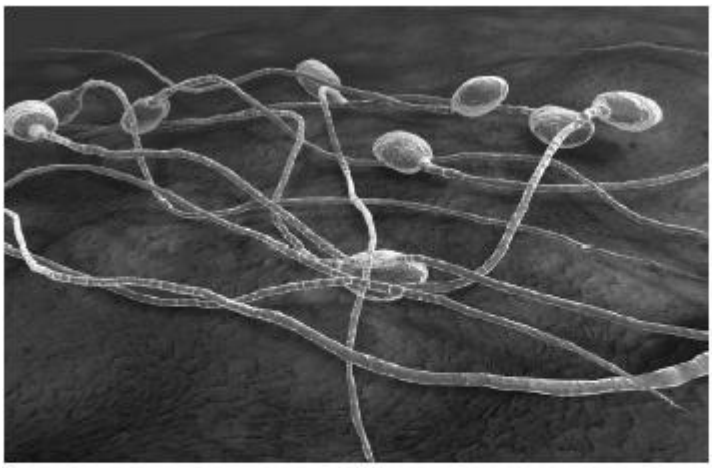
- ☒ **A** 4.2×10^{-3} μm^3
- ☒ **B** 4.2×10^{3} μm^3
- ☒ **C** 3.4×10^{-2} μm^3
- ☒ **D** 3.4×10^{2} μm^3

(c) Gamete cells are specialised for their function.

(i) The photograph shows some sperm cells, as seen using a microscope.

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×4000

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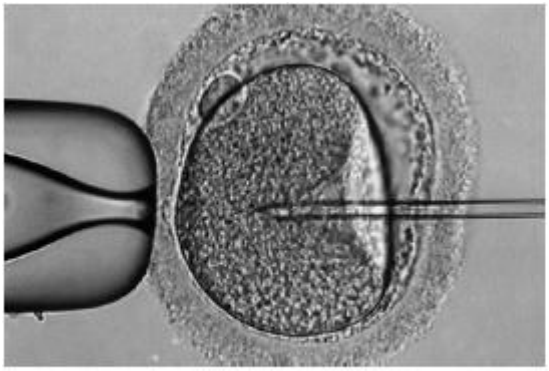
(Source: © MedicalRF.com/Alamy Stock Photo)

Which of the following statements about human sperm cells are correct?
1. they contain the same quantity of DNA as a body cell
2. they contain hydrolytic enzymes
3. the release of energy from starch enables the rotation of the flagellum

(1)

- ☒ **A** 1, 2 and 3
- ☒ **B** 2 and 3 only
- ☒ **C** 2 only
- ☒ **D** 3 only

*(ii) The photograph shows a sperm cell being injected into a human egg cell during the process of in–vitro fertilisation (IVF).



(Source: © Phanie / Alamy Stock Photo)

Scientists have suggested that one cause of human infertility is the production of egg cells with a zona pellucida thickness greater than 18 μm.
The table shows the mean thickness of the zona pellucida of fertilised egg cells and unfertilised egg cells.

Type of egg cell	Mean zona pellucida thickness / μm
fertilised	16.6 ± 3.2
unfertilised	18.9 ± 4.0

Sperm cells are released from the male by ejaculation.

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Another suggested cause of infertility is a longer time between sperm cell ejaculation and the cells reaching an egg cell in the female.
Scientists wish to increase the number of successfully fertilised egg cells by IVF. They have proposed that the technique of injecting recently obtained sperm cells should be used for egg cells with a zona pellucida thicker than 18 μm.
Evaluate this proposal.
Use the information provided in the question and your own knowledge to support your answer.

(6)

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(Total for question = 12 marks)

Q41.

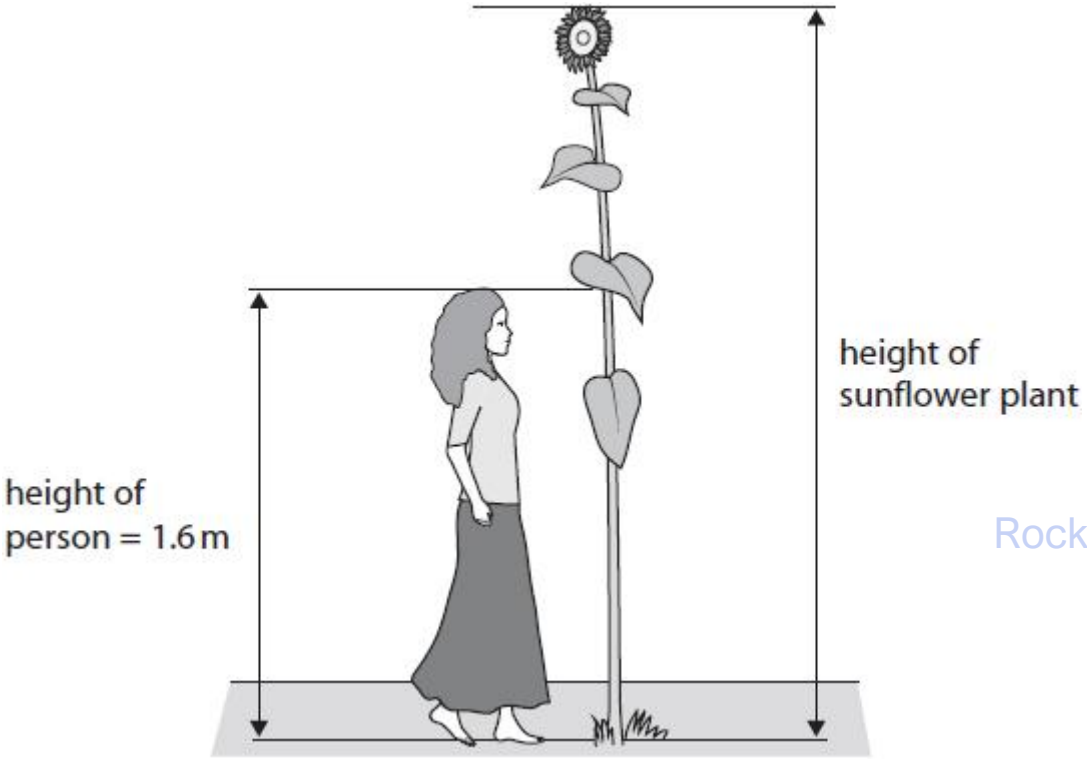
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Plants contain tissues involved in the transport of substances.

The diagram shows a sunflower plant.

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(a) Estimate the height of the sunflower, using the information in the diagram.
Give your answer with appropriate units.

(1)

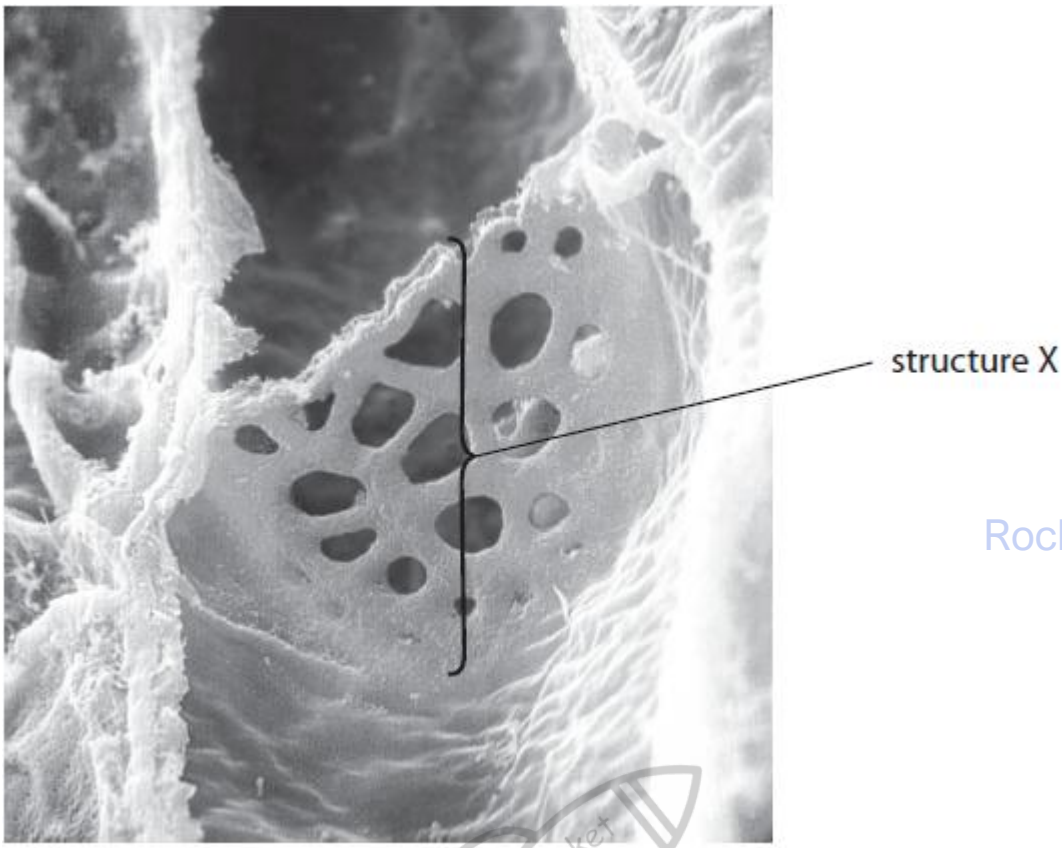
Answer .....

(b) The photograph shows a structure that would be found in a sunflower plant.

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(Source: © Dr. Richard Kessel and Dr. Gene Shih / Science Photo Library)

(i) Name the tissue that would contain structure X.

(1)

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(ii) Describe how **two** structures in this tissue enable it to carry out its function.

(2)

Structure 1 .....

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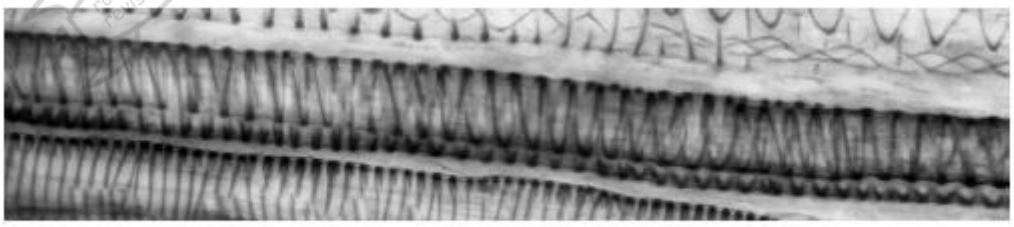
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Structure 2 .....

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(c) The photograph shows some rings of lignin, as seen using a microscope.



(Source: © Science Photo Library / Alamy Stock Photo)

(i) Which of the following tissues contain lignin?

1. phloem
2. sclerenchyma

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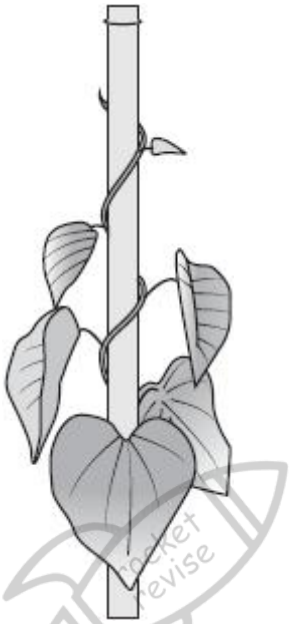
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3. xylem

(1)

- ☐ **A** 1 only
- ☐ **B** 1 and 3 only
- ☐ **C** 2 and 3 only
- ☐ **D** 3 only

(ii) The diagram shows a climbing plant that is growing up a bamboo pole.



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Suggest **two** reasons why a climbing plant may have a **lower** lignin content than a sunflower plant of the same height and diameter.

(2)

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(Total for question = 7 marks)

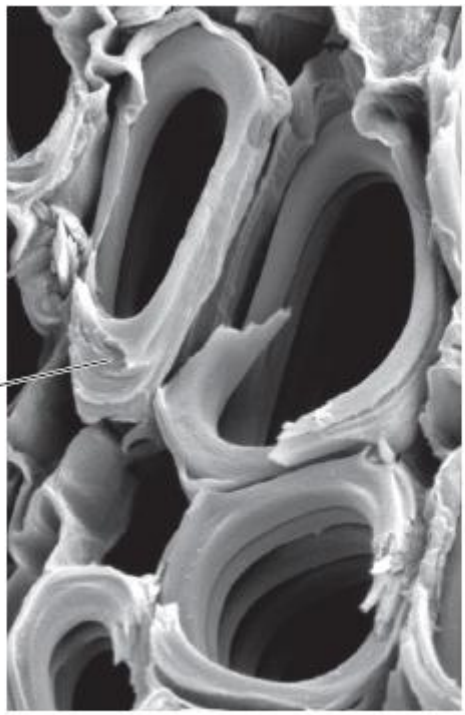
Q42.

The cells in xylem vessels have cellulose cell walls and secondary thickening.

(a) The photograph shows xylem vessels and walls with secondary thickening, as seen using an electron microscope.

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(Source: © Steve Gschmeissner / Science Photo Library)

Xylem vessels have cell walls containing cellulose microfibrils.
Describe how the structure and arrangement of cellulose **microfibrils** contribute to their functions in the cell walls of xylem vessels.

(3)

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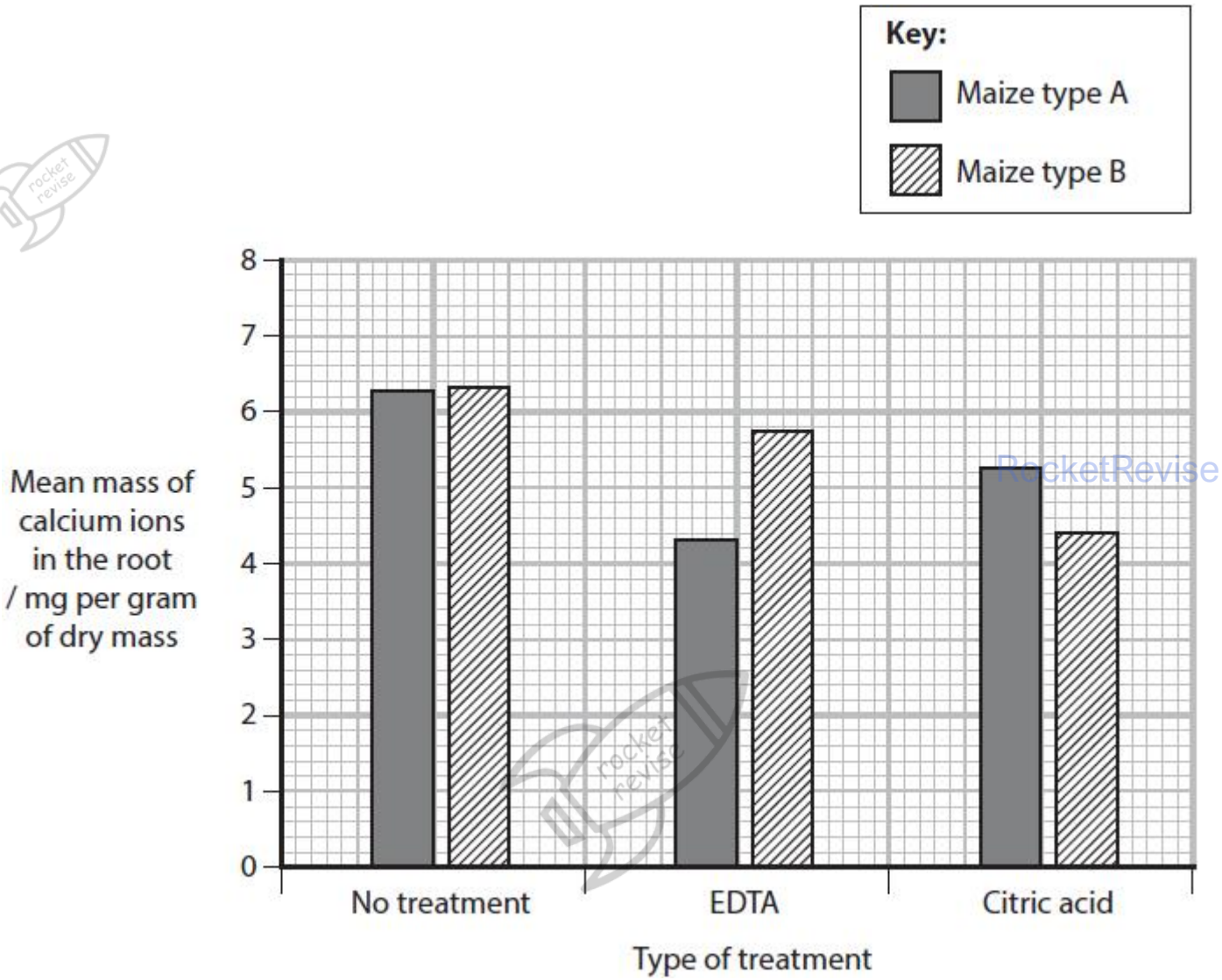
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(b) Chemicals in the soil can affect the absorption of inorganic ions.

Xylem vessels transport calcium ions from roots to other parts of a plant.
The effect of adding two chemicals, EDTA and citric acid, to the soil was investigated.
The graph shows the mean mass of calcium ions in the roots of two types of maize used in this investigation.

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(i) Give **two** conclusions for the effect of these treatments on the calcium ion content in the roots of these two types of maize.

(2)

1 .....

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2 .....

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(ii) Explain the effect EDTA would have on new cell walls in the roots of type A maize plants.

(2)

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(Total for question = 7 marks)

Q43.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Proteins can control cell processes and determine cell structure.

(a) Molecules of pre-mRNA are produced by transcription of active genes.

These molecules undergo post-transcriptional changes to form active mRNA.
Describe how post-transcriptional changes of pre-mRNA can produce active mRNA.

(3)

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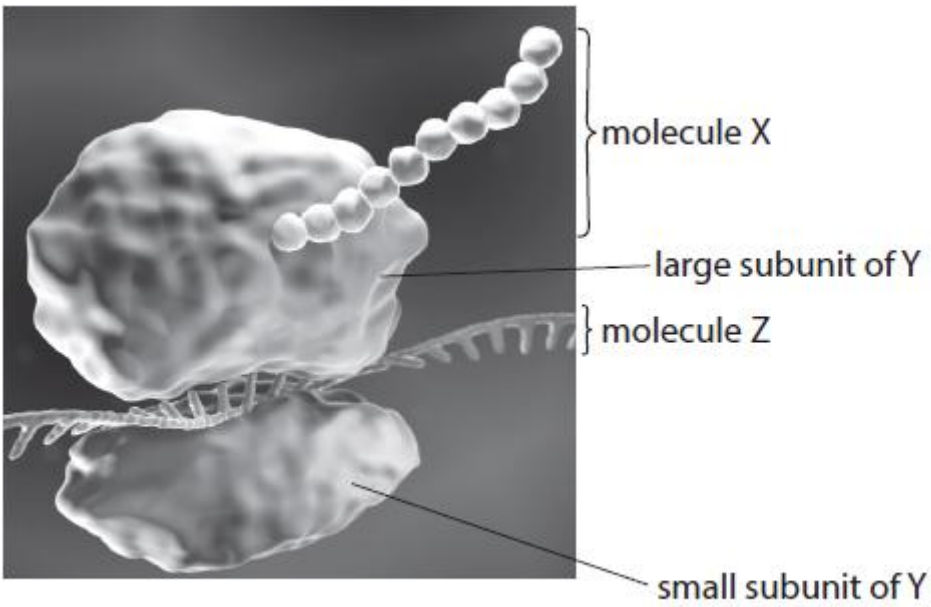
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(b) The diagram shows a process taking place on structure Y in a eukaryotic cell.

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(Source: © Artur Plawgo / Science Photo Library)

(i) Structure Y can be found in the cytoplasm or on the surface of an organelle in a eukaryotic cell. Which organelle would have structure Y on its surface?

(1)

- ☐ **A** Golgi apparatus
- ☐ **B** mitochondria
- ☐ **C** rough endoplasmic reticulum
- ☐ **D** smooth endoplasmic reticulum

(ii) Give the function of structure Y.

(1)

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(iii) Which row gives the names of the molecules labelled X and Z?

(1)

	Molecule X	Molecule Z
☐ **A**	polypeptide	DNA strand
☐ **B**	polypeptide	RNA strand
☐ **C**	polysaccharide	DNA strand
☐ **D**	polysaccharide	RNA strand

(c) DNA methylation can alter the activation of a gene.

(i) Suggest the role of the enzyme DNA methyltransferase (DNMT) in the process of DNA methylation.

(2)

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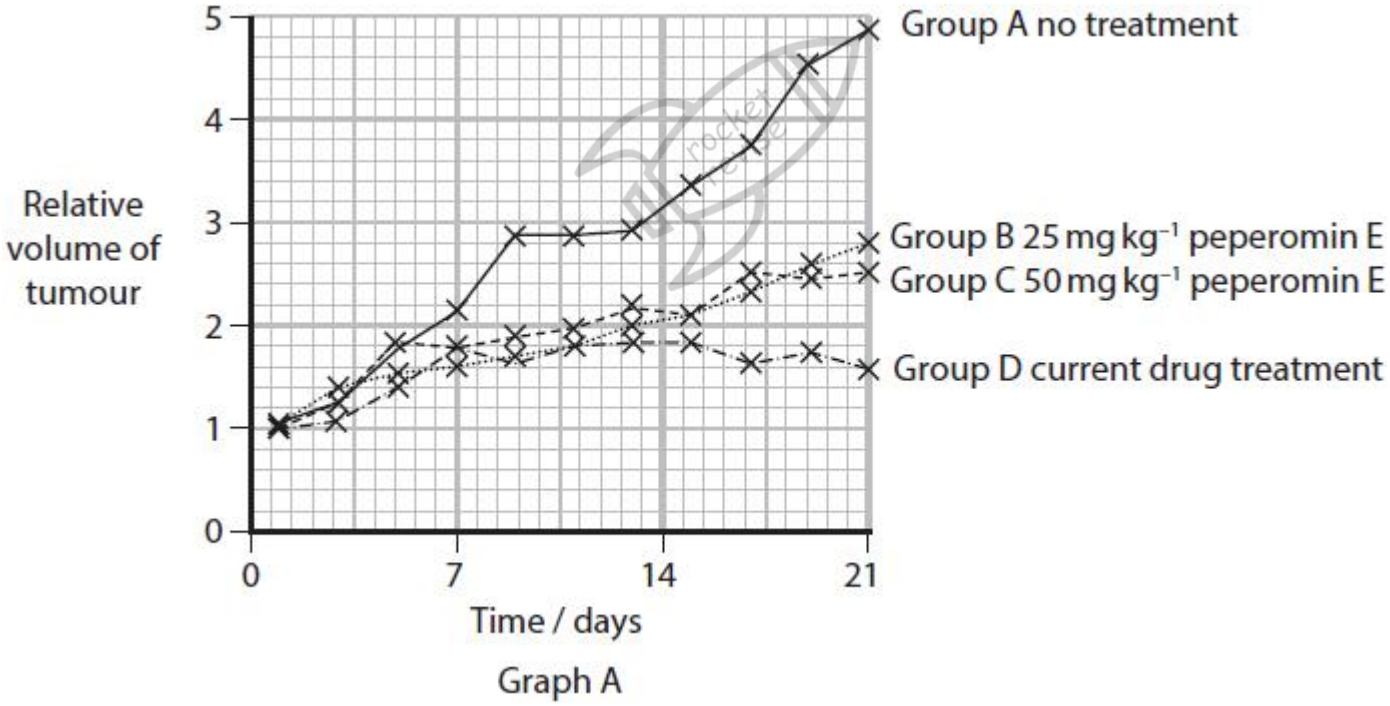
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*(ii) Some cancers can be caused by the methylation of a tumour suppressor gene called *RASSF1*.
Tumour suppressor genes can slow down cell division or cause cell death.
Peperomin E is a drug that can cause the demethylation of the gene *RASSF1*.
In an investigation different treatments were given to groups of 10 mice.
These mice had identical lung cancer tumours at the start of the investigation.
Graph A shows the changes in tumour volume for each group, over 21 days.



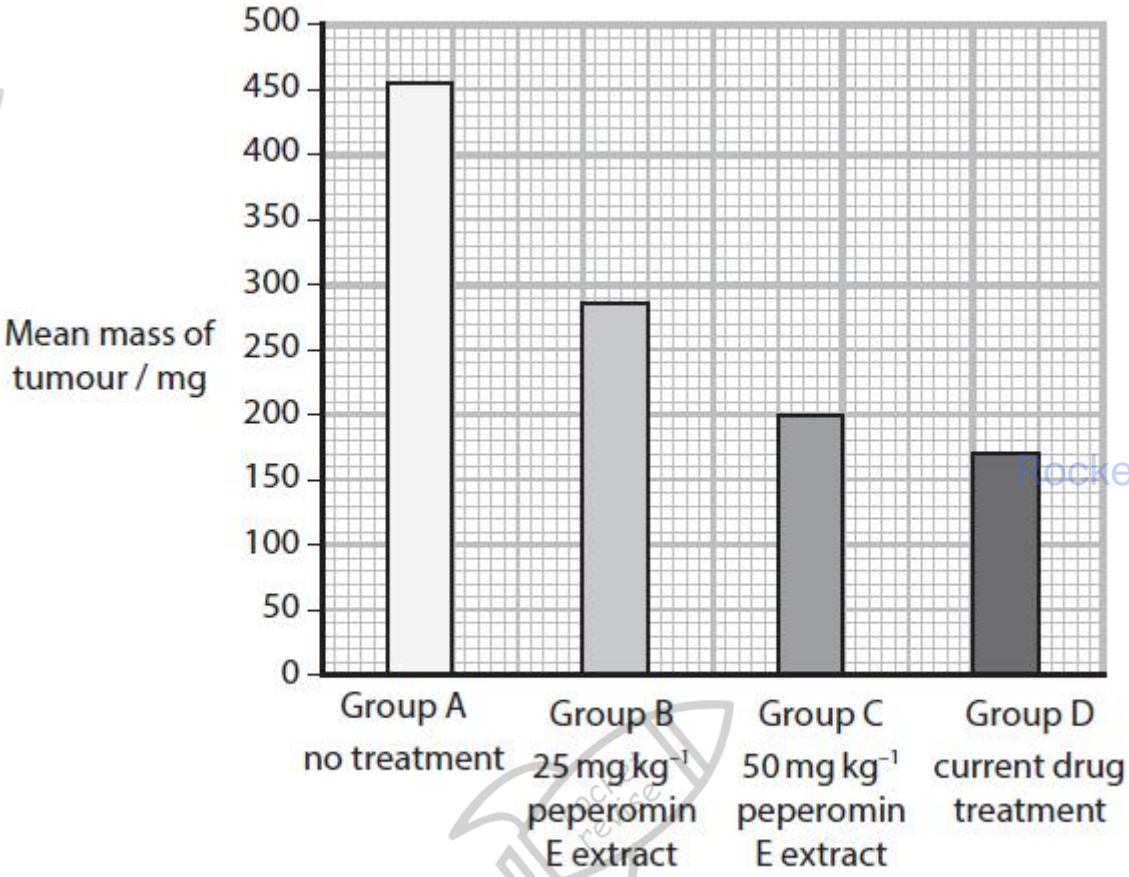
Graph A

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Graph B shows the mean mass of tumours at 21 days.



Graph B

Explain how treatment with these two drugs affects the activity of the *RASSF1* gene and the changes in tumours.
Use **all** of the information in part (c) and your own knowledge to support your answer.

(6)

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(Total for question = 14 marks)

Q44.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Parts of a bamboo plant are useful building materials due to their high tensile strength.

The photograph shows a bridge in Cambodia that has been made from parts of bamboo plants.



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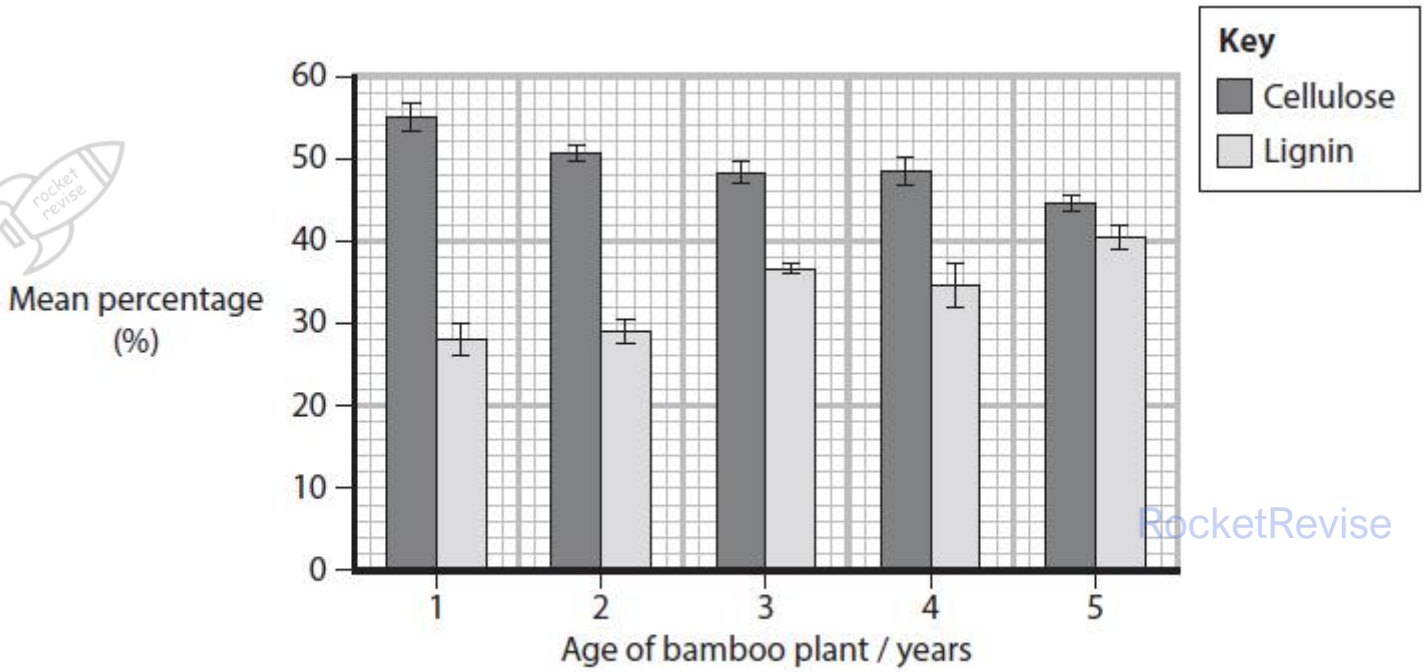
(Source: © Bjorn Svensson / Science Photo Library)

(a) Cellulose and lignin contribute to the high tensile strength of bamboo.

The graph shows the mean percentage of cellulose and lignin in bamboo plants of different ages.

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(Source: adapted from https://www.mdpi.com/2073-4360/12/1/187)

(i) Calculate the ratio of cellulose to lignin for a one-year-old bamboo plant.

(1)

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(ii) Give **one** conclusion that can be made about the effect of age on the mean percentage of **both cellulose and lignin** content of bamboo plants.

(1)

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(iii) Explain how the arrangement of molecules in the cell walls of xylem contributes to the high tensile strength of the bamboo plant.

(3)

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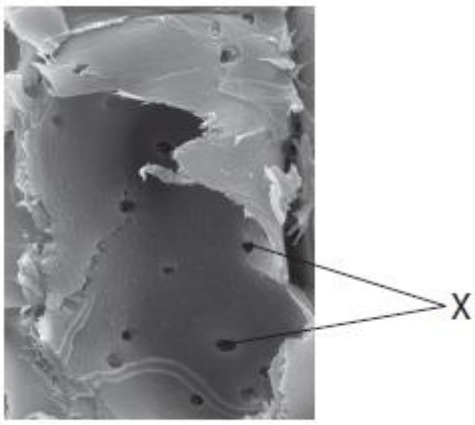
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(b) The photograph shows part of the plant cell wall from a bamboo plant.



(Source: © Eye of Science / Science Photo Library)

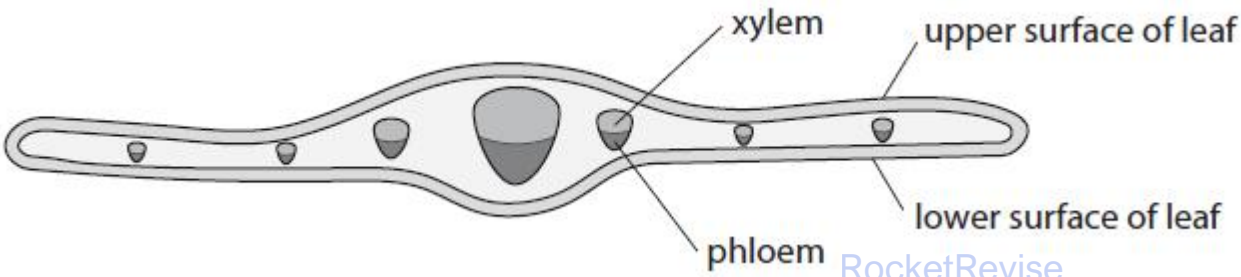
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Which are the structures labelled X?

(1)

- ☒ **A** amyloplasts
- ☒ **B** chloroplasts
- ☒ **C** plasmodesmata
- ☒ **D** tonoplasts

(c) The diagram shows the arrangement of xylem and phloem in a transverse section of a plant leaf.



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(i) Which of the following are functions of xylem vessels?

1. give support to the plant
2. translocation of organic solutes to the leaves
3. transport of inorganic ions to the leaves
4. transport of water molecules to the leaves

(1)

- ☒ **A** 1, 2 and 3 only
- ☒ **B** 1, 3 and 4 only
- ☒ **C** 1 and 3 only
- ☒ **D** 1 and 4 only

(ii) Give **one** difference between the distribution of phloem and xylem in the leaf compared with their distribution in a transverse section of a stem.

(1)

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(Total for question = 8 marks)

Q45.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Many useful products can be made from plants.

(a) The photograph shows a drinking cup.



(Source: © Alex Hinds / Alamy Stock Photo)

This cup is made from parts of a sugarcane plant that are left over from commercial production of sugar.
Explain why **this** cup would be considered a sustainable product.

(2)

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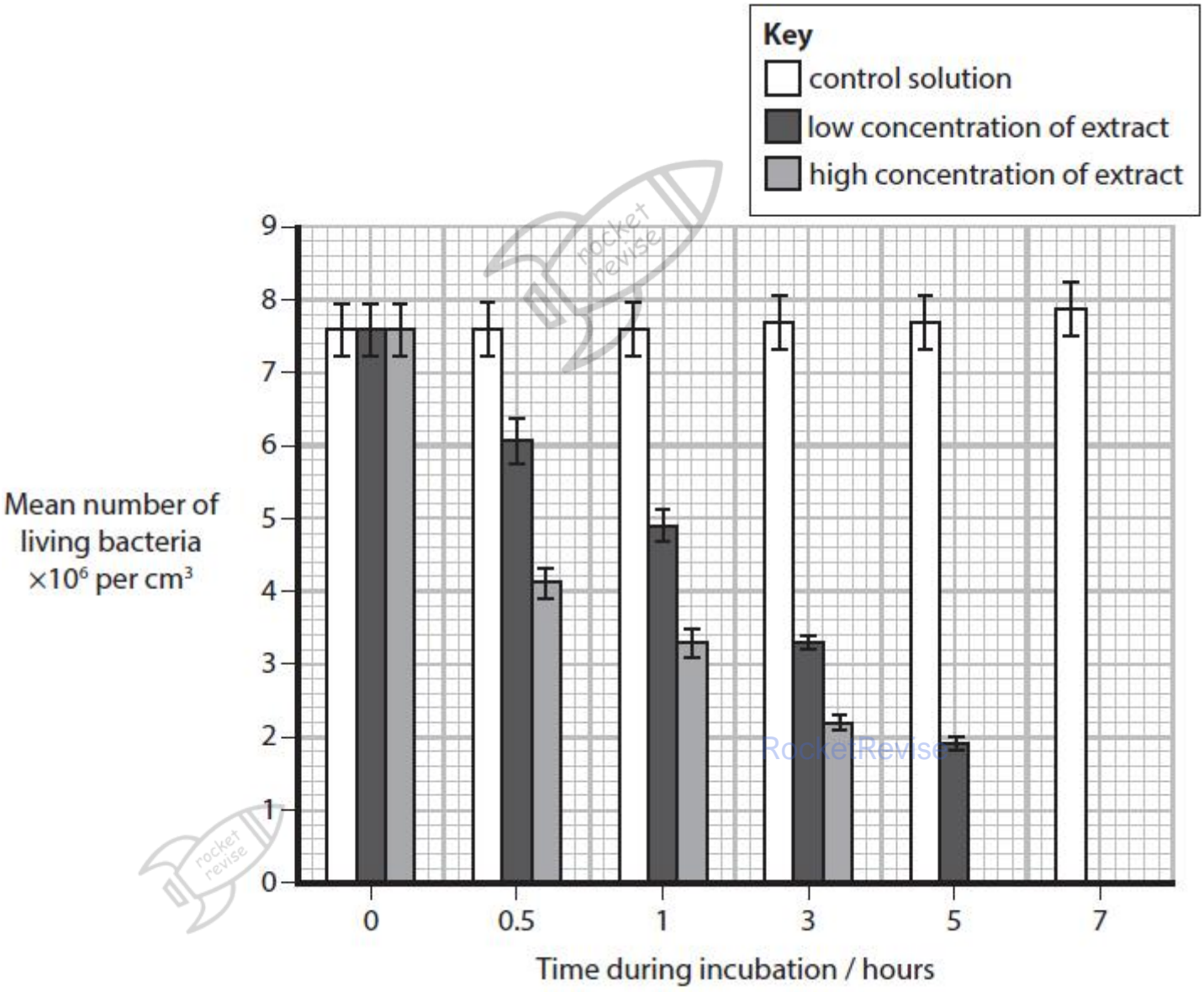
(b) Substances derived from plants can have antimicrobial properties.

The antimicrobial properties of an extract made from olive oil were investigated.
The same number of bacteria were added to tubes containing the same volume of one of the following:

- control solution
- low concentration of extract
- high concentration of extract.

A sample of bacteria was taken from each tube during incubation.
The mean number of living bacteria was determined.
The graph shows the results of this investigation.

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(Source: adapted from https://www.frontiersin.org/articles/10.3389/fmicb.2019.01586/full)

Comment on the results of this investigation.

(4)

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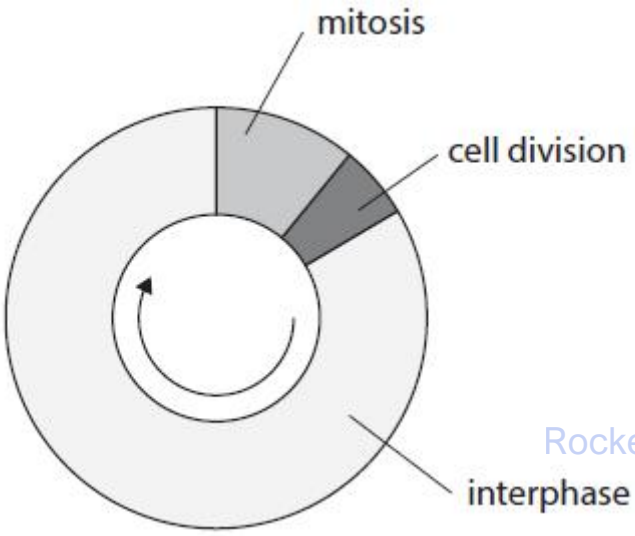
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(c) Phytohemagglutinin (PHA) is a type of molecule that can be extracted from the common bean.

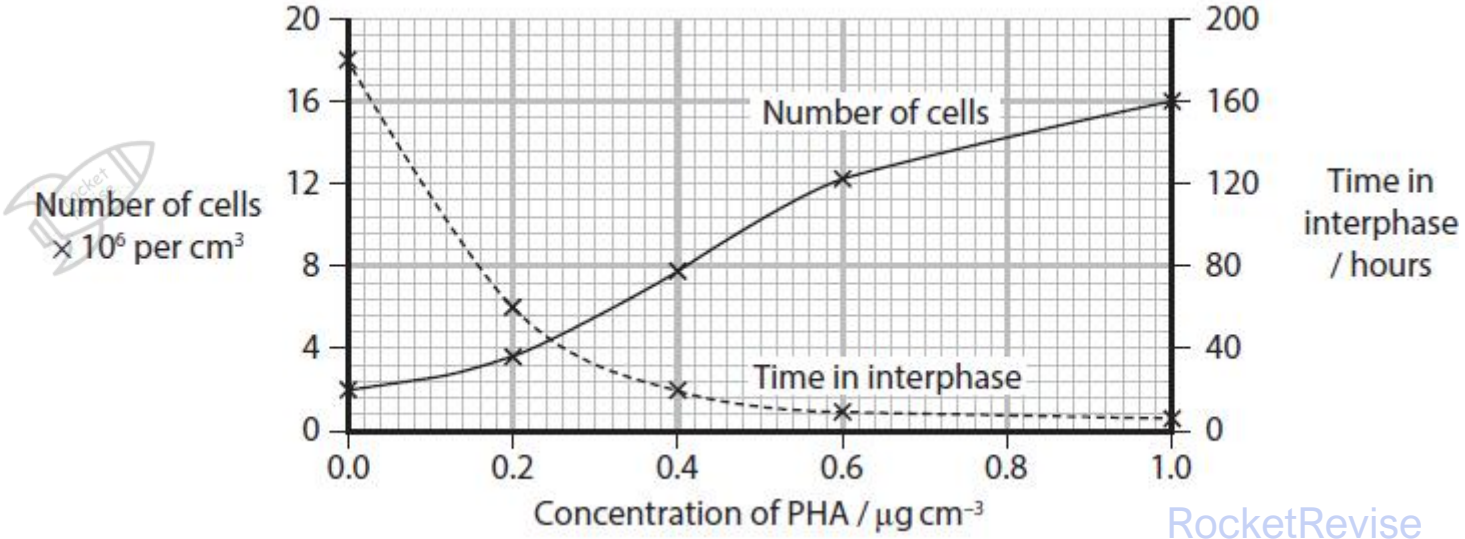
This molecule has an effect on the interphase stage of the cell cycle shown in the diagram.



In an investigation, white blood cells were treated with different concentrations of PHA.
The graph shows the effect of this treatment on the number of white blood cells and the length of time they were in interphase.

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(i) State the relationship between the time in interphase and the number of cells.

(1)

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(ii) Which is the correct percentage change in cell number from 0.0 to 1.0 µg cm^{-3} of PHA?

(1)

- ☒ **A** –700 %
- ☒ **B** –87.5 %
- ☒ **C** +87.5 %
- ☒ **D** +700 %

(iii) Explain the results of this investigation.

(3)

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(Total for question = 11 marks)

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Q46.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows strawberry plants growing in a hydroponic system.

In these systems, the plants grow in nutrient solutions instead of in soil.



(Source: © Stephanie Jackson - Agriculture / Alamy Stock Photo)

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The nutrient solutions contain inorganic ions and water.

(a) (i) Give **two** reasons why water is important to the strawberry plants.

(2)

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(ii) The nutrient solution contains nitrate ions and calcium ions. Explain the importance of these ions to the strawberry plants.

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(3)

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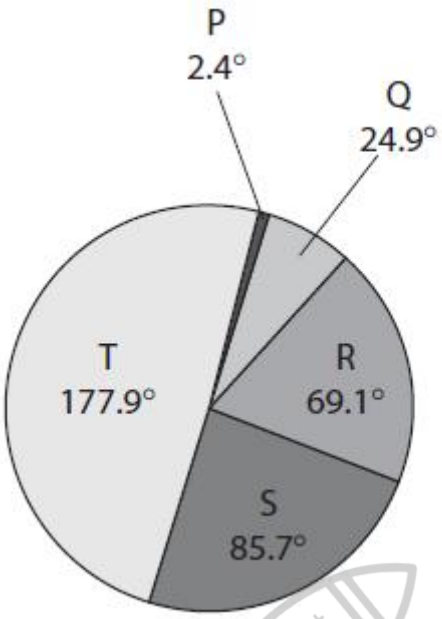
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(b) The diagram shows the strawberry production in five continents, P, Q, R, S and T, in 2021.



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Continent Q produced 634 288 000 kg of strawberries in 2021.
Which is the mass of strawberries produced in continent S?

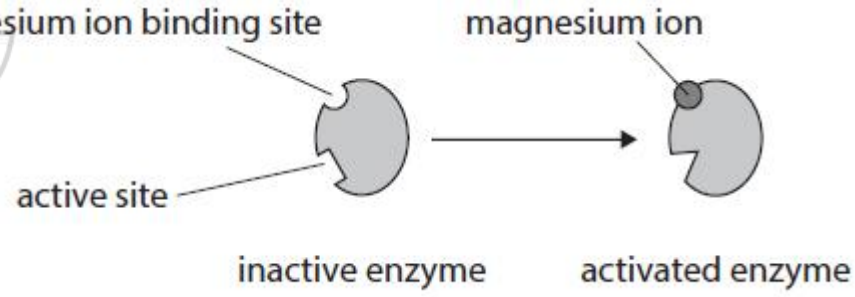
(1)

- ☒ **A** 61 136 193 kg
- ☒ **B** 760 212 883 kg
- ☒ **C** 2 183 071 550 kg
- ☒ **D** 4 531 720 288 kg

(c) In plants, magnesium ions are needed to activate enzymes required for:

- inorganic ion uptake
- protein synthesis
- photosynthesis.

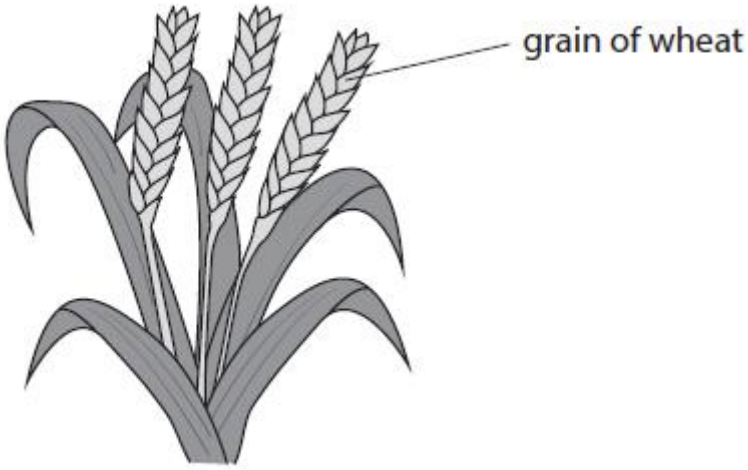
The diagram shows how the attachment of a magnesium ion activates an enzyme.



The diagram shows grains of wheat on a wheat plant.

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Grains of wheat can be ground into flour.
Flour with a higher protein content is generally more expensive.
One protein that is found in the grains of wheat plants is gluten.
The effect of magnesium ions on the mean mass of wheat grains and the mean gluten content of wheat grains was investigated.
Three groups of wheat plants were grown in separate hydroponic systems containing a nutrient solution.
The table shows the different treatments.

Group	Type of nutrient solution	Magnesium ion solution applied to leaves
A	all inorganic ions except magnesium ions	no
B	all inorganic ions except magnesium ions	yes
C	all inorganic ions including magnesium ions	no

The table shows the effect of these treatments on the mean mass of wheat grains per square metre of plants and the mean mass of gluten per kilogram of wheat grains.

Group	Mean mass of wheat grains / kg m^{-2}	Mean mass of gluten in wheat grains / g kg^{-1}
A	0.58	265
B	0.65	283
C	0.67	298

(i) Calculate the percentage difference in the mean mass of wheat grains of the plants in group A and group B.
Use the equation:

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percentage difference between A and B $= \dfrac{A - B}{\left(\dfrac{A + B}{2}\right)} \times 100$

Give your answer to an **appropriate number** of decimal places.

(2)

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Answer .....%

*(ii) Farmers concluded that they would make more money if magnesium ions were added to growing wheat plants.

Discuss this conclusion.

Use **all** the information in part (c) to support your answer.

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(Total for question = 14 marks)

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